	Course Name: Advanced Web Technology	EXPERIMENT NO. 1	
	Course Code: 20CP314P Faculty: Komal Singh	Branch: CSE	Semester: VI
Submitted by: Harsh Shah Roll no: 21BCP359			

Objective: Building a Simple Web Server in Node.js

Experiment: Implement the NodeJS Programs.

Q.1 Write Node JS programs for following modules:

a. http

```
const http = require('http');
const server = http.createServer((req, res) =>
{ res.writeHead(200, { 'Content-Type': 'text/plain' });
res.end('Hello World!\n');
});
server.listen(3000, 'localhost', () =>
{ console.log('Server running at
http://localhost:3000/')
})
```

Output:-

```
Hello World!
```

```
[nodemon] restarting due to changes...
[nodemon] starting `node temp.js`
Server running at http://localhost:3000/
```

b. filesystem

```
const fs = require('fs');
const fileName = 'example.txt';
const fileContent = 'This is a sample text file.';
fs.writeFile(fileName, fileContent, (err) => {
  if (err) throw err;
  console.log(`${fileName} has been created.`);
});
fs.readFile(fileName, 'utf8', (err, data) => {
  if (err) throw err;
  console.log(`Contents of ${fileName}:`);
  console.log(data);
});
```

Output:-

```
example.txt has been created.
Contents of example.txt:
This is a sample text file.
```

c. URL

```
const url = require('url');
const urlString =
  'https://www.example.com/path/to/resource?query1=value1&query2=value2';
const parsedUrl = url.parse(urlString, true);
console.log('Parsed URL:');
console.log(parsedUrl);
```

Output:-

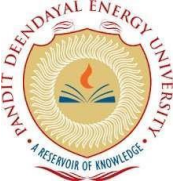
```
Parsed URL:
Url {
  protocol: 'https:',
  slashes: true,
  auth: null,
  host: 'www.example.com',
  port: null,
  hostname: 'www.example.com',
  hash: null,
  search: '?query1=value1&query2=value2',
  query: [Object: null prototype] { query1: 'value1', query2: 'value2' },
  pathname: '/path/to/resource',
  path: '/path/to/resource?query1=value1&query2=value2',
  href: 'https://www.example.com/path/to/resource?query1=value1&query2=value2'
}
```

d. Event

```
const EventEmitter = require('events');
class MyEmitter extends EventEmitter {}
const myEmitter = new MyEmitter();
myEmitter.on('event', () => {
  console.log('Event emitted!');
});
myEmitter.emit('event');
```

Output:-

```
[nodemon] starting node temp.js
Event emitted!
[nodemon] clean exit - waiting for changes
```

	Course Name: Advanced Web Technology	EXPERIMENT NO. 2	
	Course Code: 20CP314P Faculty: Komal Singh	Branch: CSE	Semester: VI
Submitted by: Harsh Shah Roll no: 21BCP359			

Objective: Building a Web Server with Express.js

Experiment 2: Implement with routing methods (GET and POST) and also implement one built-in middleware, configurable middleware and one third party middleware (Cookie-Parser).

Hint: You can also use the function validate cookie in third party middleware.

App.js

```
const express = require('express');
const cookieParser = require('cookie-parser'); const app = express();
app.use(express.json());
app.use(express.urlencoded({ extended: true })); app.use(cookieParser());

// Configurable middleware
const myMiddleware = (config)
=> { return (req, res, next)
=> {
  console.log(`Configurable Middleware:
    ${config}`); next();
};
};

// GET route
app.get('/', (req, res) =>
{ res.send('Hello, this is a GET
  route!');
});

// POST route with configurable middleware and cookie handling
app.post('/', myMiddleware('I am configurable middleware'), (req, res) =>
{
  console.log('Request Body: ', req.body);
});
```

```
// Retrieve and validate the cookie
const myCookie =
req.cookies.myCookie; if (myCookie
=== 'cookieValue') {
    res.send('Hello, this is a POST route! Cookie is valid.');
```

```
  } else {
    res.send('Hello, this is a POST route! Cookie is invalid.');
```

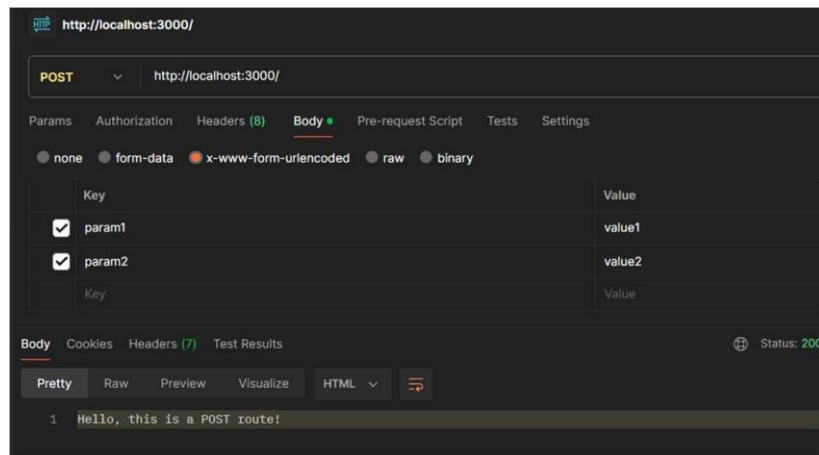
```
  }
});
const PORT = 3000;
app.listen(PORT, ()
=> {
    console.log(`Server is running on http://localhost:${PORT}`);
```

Output :-

Get Request :-



Post Request With Cokkie-parser :-



	Course Name: Advanced Web Technology	EXPERIMENT NO. 3	
	Course Code: 20CP314P Faculty: Komal Singh	Branch: CSE	Semester: VI
(To be filled by Student) Submitted by: Harsh Shah Roll no: 21BCP359			

Objective: Setting up a MongoDB Database (Connecting MongoDB to your application)

Experiment 3: Create a JavaScript file with MongoDB queries for operations such as insert, update, and delete while also establishing a connection to the MongoDB database.

Hint: Ensure that your MongoDB server is running and accessible at localhost:27017 or replace it with the appropriate connection string if it's hosted elsewhere.

Code:-

```
const mongoose = require('mongoose');
```

```
const validator = require('validator');
```

```
const uri =
```

```
'mongodb+srv://harshshah:harsh@clusterO.Imhkvus.mongodb.net/harsh?retryWrites=true&w=majority';
```

```
mongoose.connect(uri, { useNewUrlParser: true, useUnifiedTopology: true });
```

```
const friendSchema = new mongoose.Schema({
```

```
  name: String,
```

```
  age: Number,
```

```
  email: {
```

```
    type: String,
```

```
    required: true,
```

```
    validate: {
```

```
      validator: validator.isEmail,
```

```
        message: '{VALUE} is not a valid email'
    }
},
active: Boolean
});
```

```
const Friend = mongoose.model("Harsh_W", friendSchema);
const inst_v = [
    { name: "HarshShah", age: 20, email: "HarshShah@gmail.com", active: true },
    { name: "Karan", age: 45, email: "Karan@example.com", active: true } // Added comma
    after age property
];
```

```
async function main() {
    try {
        // Insert documents
        await Friend.insertMany(inst_v);

        // Update documents
        await Friend.updateMany({ name: "Karan" }, { $set: { age: 45 } });

        // Delete documents
        await Friend.deleteOne({ name: "Karan" });

        // Read documents
        const read = await Friend.find({ name: "HarshShah" });
        console.log(read);

    } catch (err) {
        console.log(err);
    } finally {
```

```
        mongoose.connection.close();
    }
}
main()
```

Output:-

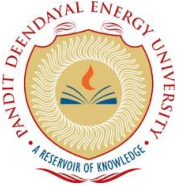
1.) INSERT

```
[
  {
    _id: ObjectId("65c6e1b53ea5e8f810f5e3d1"),
    name: "HarshShah",
    email: "HarshShah@gmail.com",
    age: 20,
    active: true
  },
  {
    _id: ObjectId("65c6e1b53ea5e8f810f5e3d2"),
    name: "Karan",
    email: "Karan@example.com",
    age: 45,
    active: true
  }
]
```

2.) UPDATE AND DELETE

```
[
  {
    _id: ObjectId("65c6e1b53ea5e8f810f5e3d1"),
    name: "HarshShah",
```

```
    email: "HarshShah@gmail.com",  
    age: 20,  
    active: true  
  }  
]
```


	Course Name: Advanced Web Technology	EXPERIMENT NO. 4
	Course Code: 20CP314P	Branch: CSE
	Faculty: Komal Singh	Semester: VI
Submitted by: Harsh Shah Roll no: 21BCP359		

Objective: Building models with Mongoose

Create a database schema and model using mongoose and perform MongoDB queries for operations such as insert, update, and delete while also establishing a connection to the MongoDB database.

Mongoose Connection

```
const mongoose = require("mongoose");

mongoose
  .connect("mongodb://localhost/mydb")
  .then(() => console.log("Connected to MongoDB successfully!"))
  .catch((error) => console.error("Error connecting to MongoDB", error));
```

```
(base) PS D:\DEV\Web Development\PDEU-AWT\Practical_4> npm start

> practical_4@1.0.0 start
> nodemon task1.js

[nodemon] 3.1.0
[nodemon] to restart at any time, enter `rs`
[nodemon] watching path(s): *.*
[nodemon] watching extensions: js,mjs,cjs,json
[nodemon] starting `node task1.js`
Connected to MongoDB successfully!
```

Database Schema

```
const schema = new mongoose.Schema({

  name: {
    type: String,
    required: true,
  },
  email: {
    type: String,
```

```
    required: true,
    unique: true,
  },
  age: {
    type: Number,
    default: 0,
  },
  deregistered: {
    type: Date,
    default: Date.now,
  },
});
const obj = mongoose.model("subsription", schema);
```

Insert Operation

```
const newobj = new obj({
  name: "Harsh Shah",
  email: "harsh.sce21@sot.pdpu.ac.in",
  age: 20,
});
newobj
  .save()
  .then((kri) => console.log("User created:", kri))
  .catch((err) => console.error(err));
```

```
Connected to MongoDB successfully!
User created: {
  name: 'Harsh Shah',
  email: 'harsh.sce21@sot.pdpu.ac.in',
  age: 20,
  _id: new ObjectId('660fd63ed60d5b00ccc41371'),
  deregistered: 2024-04-05T10:45:18.554Z,
  __v: 0
}
```

```
_id: ObjectId('660fd63ed60d5b00ccc41371')
name : "Harsh Shah"
email : "harsh.sce21@sot.pdpu.ac.in"
age : 20
deregistered : 2024-04-05T10:45:18.554+00:00
__v : 0
```

a. Insert One

```
const Newuser = new obj({ name: "krish", email: "ky@gmail.com", age: 20 });
```

```
obj.collection
```

```
.insertOne(Newuser)
```

```
.then((data) => {
```

```
  console.log(data);
```

```
})
```

```
.catch((err) => {
```

```
  console.log(err);
```

```
});
```

```
Connected to MongoDB successfully!
{
  acknowledged: true,
  insertedId: new ObjectId('660fd845ddcedbff28571b2e')
}
```

b. Insert Many

```
const Newuser = new obj({ name: "krishna", email: "ku@gmail.com", age: 20 });
```

```
const Newuser1 = new obj({ name: "kriti", email: "kr@gmail.com", age: 21 });
```

```
obj.collection
```

```
.insertMany([Newuser, Newuser1])
```

```
.then((data) => {
```

```
  console.log(data);
```

```
})
```

```
.catch((err) => {
```

```
  console.log(err);
```

```
});
```

```
Connected to MongoDB successfully!
{
  acknowledged: true,
  insertedCount: 2,
  insertedIds: {
    '0': new ObjectId('660fd8829d209dcbb3368d5b'),
    '1': new ObjectId('660fd8829d209dcbb3368d5c')
  }
}
```

Update Operation

a. Update One

```
obj.updateOne({ name: "krishna" }, { $set: { name: "madhav" } })  
  .then((docs) => {  
    if (docs) {  
      console.log(docs);  
    } else {  
      console.log("no such user exist");  
    }  
  })  
  .catch((err) => {  
    console.log(err);  
  });
```

```
Connected to MongoDB successfully!  
{  
  acknowledged: true,  
  modifiedCount: 1,  
  upsertedId: null,  
  upsertedCount: 0,  
  matchedCount: 1  
}
```

b. Update Many

```
obj.updateMany({age: 20},{ $set:{ age: 15}})  
  .then((docs)=>{  
    if(docs) {  
      console.log(docs);  
    } else {  
      console.log("no such user exist");  
    }  
  })  
  .catch((err)=>{  
    console.log(err);  
  });
```

```
Connected to MongoDB successfully!  
{  
  acknowledged: true,  
  modifiedCount: 4,  
  upsertedId: null,  
  upsertedCount: 0,  
  matchedCount: 4  
}
```

Find/Search Operation

a. Find without query

```
obj.find({})  
  
  .then((doc) => {  
    console.log(doc);  
  })  
  
  .catch((err) => {  
    console.log(err);  
  });
```

```
Connected to MongoDB successfully!  
[  
  {  
    _id: new ObjectId('660fd63ed60d5b00ccc41371'),  
    name: 'Harsh Shah',  
    email: 'harsh.sce21@sot.pdpu.ac.in',  
    age: 15,  
    deregistered: 2024-04-05T10:45:18.554Z,  
    __v: 0  
  },  
  {  
    _id: new ObjectId('660fd845ddcedbff28571b2e'),  
    name: 'krish',  
    email: 'ky@gmail.com',  
    age: 15,  
    deregistered: 2024-04-05T10:53:57.063Z  
  },  
  {  
    _id: new ObjectId('660fd8829d209dcbb3368d5b'),  
    name: 'madhav',  
    email: 'ku@gmail.com',  
    age: 15,  
    deregistered: 2024-04-05T10:54:58.784Z  
  },  
  {  
    _id: new ObjectId('660fd8829d209dcbb3368d5c'),  
    name: 'kriti',  
    email: 'kr@gmail.com',  
    age: 21,  
    deregistered: 2024-04-05T10:54:58.784Z  
  },  
  {  
    _id: new ObjectId('660fd92a27c646b3dd442ec0'),  
    name: 'dev',  
    email: 'dev@gmail.com',  
    age: 15,  
    deregistered: 2024-04-05T10:57:46.699Z  
  }  
]
```

b. Find with query

```
obj.find({ name: "dev" })  
  
  .then((doc) => {  
    console.log(doc);  
  })  
  
  .catch((err) => {  
    console.log(err);  
  });
```

```
Connected to MongoDB successfully!
[
  {
    _id: new ObjectId('660fd92a27c646b3dd442ec0'),
    name: 'dev',
    email: 'dev@gmail.com',
    age: 15,
    deregistered: 2024-04-05T10:57:46.699Z
  }
]
```

Delete Operation

a. Delete One

```
obj.deleteOne({name: "kriti"})
.then((docs)=>{
  if(docs) {
    console.log(docs);
  } else {
    console.log("no such user exist");
  }
}).catch((err)=>{
  console.log(err);
});
```

```
Connected to MongoDB successfully!
{ acknowledged: true, deletedCount: 1 }
```

b. Delete Many

```
obj.deleteMany({age: 21})
.then((docs)=>{
  if(docs) {
    console.log(docs);
  } else {
    console.log("no such user exist");
  }
}).catch((err)=>{
  console.log(err);
});
```

```
Connected to MongoDB successfully!
{ acknowledged: true, deletedCount: 4 }
```

MongoDB Compass

mydb.subscriptions

4DOCUMENTS2INDEXES

Documents

Aggregations

Schema

Indexes

Validation

Filter

Type a query: { field: 'value' } or [Generate query](#)

Explain

Reset

Find

</>

Options

ADD DATA

EXPORT DATA

UPDATE

DELETE

1 - 4 of 4

```
{
  "_id": ObjectId('660fd63ed60d5b00ccc41371'),
  "name": "Harsh Shah",
  "email": "harsh.sce21@sot.pdpu.ac.in",
  "age": 15,
  "deregistered": 2024-04-05T10:45:18.554+00:00,
  "__v": 0
}
```

```
{
  "_id": ObjectId('660fd845ddcedbff28571b2e'),
  "name": "krish",
  "email": "ky@gmail.com",
  "age": 15,
  "deregistered": 2024-04-05T10:53:57.063+00:00
}
```

```
{
  "_id": ObjectId('660fd8829d209dcbb3368d5b'),
  "name": "madhav",
  "email": "ku@gmail.com",
  "age": 15,
  "deregistered": 2024-04-05T10:54:58.784+00:00
}
```

My Queries

mydb

mydb

subscriptions

mydb.subscriptions

0DOCUMENTS2INDEXES

Documents

Aggregations

Schema

Indexes

Validation

Filter

Type a query: { field: 'value' } or [Generate query](#)

Explain

Reset

Find

</>

Options

ADD DATA

EXPORT DATA

UPDATE

DELETE

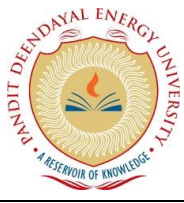
0 - 0 of 0



This collection has no data

It only takes a few seconds to import data from a JSON or CSV file.

Import Data

	Course Name: Advanced Web Technology	EXPERIMENT NO. 5
	Course Code: 20CP314P	Branch: CSE
	Faculty: Komal Singh	Semester: VI
Submitted by: Harsh Shah Roll no: 21BCP359		

Objective: Create your first React code

1. Create a “hello world” react app.

index.js

```
import React from "react";
import ReactDOM from "react-dom/client";
import "./index.css";
import App from "./App";
import reportWebVitals from "./reportWebVitals";

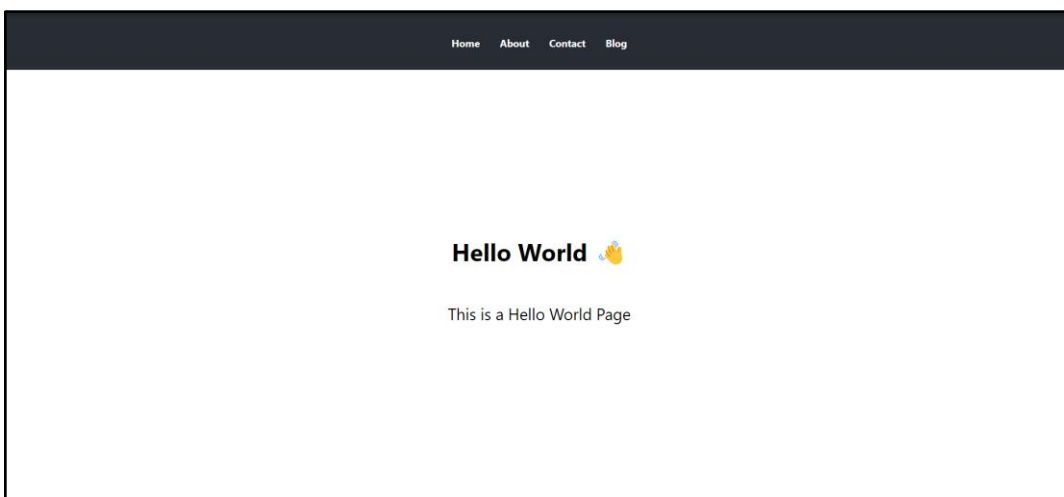
const root = ReactDOM.createRoot(document.getElementById("root"));
root.render(
  <React.StrictMode>
    <App />
  </React.StrictMode>
);
reportWebVitals();
```

App.js

```
import "./App.css";
import Home from "./components/Home";
import { BrowserRouter, Routes, Route } from "react-router-dom";
function App() {
  return (
    <div>
      <BrowserRouter>
        <Routes>
          <Route path="/" element={<Home />}>
            {" "}
          </Route>
        </Routes>
      </BrowserRouter>
    </div>
  );
}
export default App;
```

Home.jsx

```
import React from "react";
const Home = () => {
  return
    <div>
      <header className="App-header">
        <h2>Hello World 🙌 </h2>
        <p>This is a Hello World Page</p>
      </header>
    </div>
  );
};
export default Home;
```

Output**2. Create a multiple web page commercial furniture application having a navigation bar using react components having pages such as home, products and customer review.****index.js**

```
import React from 'react';
import ReactDOM from 'react-dom/client';
import './index.css';
import './App.css';
import App from './App';
import reportWebVitals from './reportWebVitals';

const root = ReactDOM.createRoot(document.getElementById('root'));
root.render(
  <React.StrictMode>
    <App/>
  </React.StrictMode>
);
reportWebVitals();
```

App.js

```
import './App.css';
import NavBar from './components/NavBar';
import Home from './components/Home';
import Products from './components/Products';
import Review from './components/Review';
import { BrowserRouter, Routes, Route } from 'react-router-dom';

function App() {
  return (
    <div>
      <BrowserRouter>
        <NavBar />
        <Routes>
          <Route path="/" element={<Home />}>
            {" "}
          </Route>
          <Route path="/reviews" element={<Review />}>
            {" "}
          </Route>
          <Route path="/products" element={<Products />}>
            {" "}
          </Route>
        </Routes>
      </BrowserRouter>
    </div>
  );
}
export default App;
```

Home.jsx

```
import React from 'react';
import './Home.module.css';

const Home = () => {
  return (
    <div>
      <header className="App-header">
        <div className="home">
          <div className="hero-section">
            <h1>Welcome to Our Commercial Furniture Store</h1>
            <p>
              We provide high-quality furniture solutions for
              commercial spaces.
            </p>
          </div>
          <div className="about-section">
            <h2>About Us</h2>
            <p>
              We are dedicated to providing durable and stylish
              furniture options for businesses, offices,
              restaurants, hotels, and more.
            </p>
            <p>
              Our team is committed to delivering exceptional
```

```

        customer service and ensuring satisfaction with
        every purchase.
    </p>
</div>
<div className="contact-section">
    <h2>Contact Us</h2>
    <ul>
        <li>Email: info@furniturestore.com</li>
        <li>Phone: 123-456-7890</li>
    </ul>
</div>
</div>
</header>
</div>
);
};
export default Home;

```

Home.module.css

```

.home {
    padding: 20px;
}

.hero-section {
    text-align: center;
    margin-bottom: 50px;
}

.hero-section h1 {
    font-size: 36px;
    margin-bottom: 20px;
}

.hero-section p {
    font-size: 18px;
    margin-bottom: 30px;
}

.btn {
    display: inline-block;
    padding: 10px 20px;
    background-color: #007bff;
    color: #fff;
    text-decoration: none;
    border-radius: 5px;
    transition: background-color 0.3s ease;
}

.btn:hover {
    background-color: #0056b3;
}

.about-section,
.contact-section {
    margin-bottom: 50px;
}

.about-section h2,
.contact-section h2 {
    font-size: 24px;
    margin-bottom: 20px;
}

.about-section p,

```

```
.contact-section p {
  font-size: 16px;
  margin-bottom: 15px;
}
.contact-section ul {
  list-style: none;
  padding-left: 0;
}
.contact-section li {
  font-size: 16px;
  margin-bottom: 10px;
}
```

NavBar.jsx

```
import React from "react";
import { Link } from "react-router-dom";
import styles from "./NavBar.module.css";
const NavBar = () => {
  return (
    <nav className={styles.navBar}>
      <ul className={styles.navList}>
        <li>
          <Link to="/">Home</Link>
        </li>
        <li>
          <Link to="/products">Products</Link>
        </li>
        <li>
          <Link to="/reviews">Reviews</Link>
        </li>
      </ul>
    </nav>
  );
};
export default NavBar;
```

NavBar.module.css

```
.navBar {
  background-color: #282c34;
  padding: 1em;
}
.navList {
  list-style-type: none;
  padding: 0;
  display: flex;
  justify-content: center;
}
.navList li {
  margin: 0 1em;
}
.navList li a {
  text-decoration: none;
  color: white;
  font-weight: bold;
}
```

Furniture.js

```
import "../App.css";
function Furniture(props) {
  return (
    <div className="grid-item">
      
      <h3 className="itemName">Trotten</h3>
      <p className="itemPrice">16690</p>
    </div>
  );
}
export default Furniture;
```

Products.jsx

```
import "../App.css";
import Furniture from "../Furniture";

function Products() {
  return (
    <div className="grid-container App-header App">
      <Furniture />
      <Furniture />
      <Furniture />
      <Furniture />
      <Furniture />
      <Furniture />
      <Furniture />
      <Furniture />
      <Furniture />
    </div>
  );
}

export default Products;
```

Review.jsx

```
import React from "react";
import "../App.css";
const reviews = [
  {
    id: 1,
    name: "John Doe",
    rating: 5,
    review: "Excellent product! Very satisfied with the quality and performance.",
  },
  {
    id: 2,
    name: "Jane Smith",
    rating: 4,
```

```

    review: "Great service! Fast delivery and good customer support.",
  },
  {
    id: 3,
    name: "David Brown",
    rating: 3,
    review: "Product was okay, but could be improved. Delivery was delayed.",
  },
];

```

```

const Reviews = ({ name, rating, review }) => {
  return (
    <div className="review">
      <h3>{name}</h3>
      <div className="rating">Rating: {rating}/5</div>
      <p>{review}</p>
    </div>
  );
};

```

```

const CustomerReviews = ({ reviews }) => {
  return (
    <div className="customer-reviews">
      <h2>Customer Reviews</h2>
      {reviews.map((review) => (
        <Reviews key={review.id} {...review} />
      ))}
    </div>
  );
};

```

```

const Review = () => {
  return (
    <div>
      <header className="App-header">
        <CustomerReviews reviews={reviews} />
      </header>
    </div>
  );
};

```

```
export default Review;
```

App.css

```

.App {
  text-align: center;
  width: 97.5vw;
}
.App-logo {
  height: 40vmin;
  pointer-events: none;
}
@media (prefers-reduced-motion: no-preference) {
  .App-logo {
    animation: App-logo-spin infinite 20s linear;
  }
}

```

```
.App-header {
  background-color: #282c34;
  min-height: 100vh;
  display: flex;
  flex-direction: column;
  align-items: center;
  justify-content: center;
  font-size: calc(10px + 2vmin);
  color: white;
}
.App-link {
  color: #61dafb;
}
@keyframes App-logo-spin {
  from {
    transform: rotate(0deg);
  }
  to {
    transform: rotate(360deg);
  }
}
.link {
  color: white;
}
.grid-container {
  display: inline-grid;
  grid-template-columns: auto auto auto;
  grid-gap: 20px;
  text-align: center;
  padding: 20px;
}
.grid-item {
  background-color: #f2f2f2;
  padding: 20px;
  text-align: center;
  color: black;
  border-radius: 25px;
}
.customer-reviews {
  margin: 20px auto;
  width: 70%;
}
.review {
  border: 1px solid #ccc;
  border-radius: 8px;
  padding: 20px;
  margin-bottom: 20px;
  box-shadow: 0 0 10px rgba(0, 0, 0, 0.1);
}
.rating {
  font-weight: bold;
  margin-bottom: 10px;
}
```

Output:

Home Page

HomeProductsReviews

Welcome to Our Commercial Furniture Store

We provide high-quality furniture solutions for commercial spaces.

About Us

We are dedicated to providing durable and stylish furniture options for businesses, offices, restaurants, hotels, and more.

Our team is committed to delivering exceptional customer service and ensuring satisfaction with every purchase.

Contact Us

Email: info@furniturestore.com

Phone: 123-456-7890

Products Page

HomeProductsReviews



Trotten

16690



Trotten

16690



Trotten

16690



Trotten

16690



Trotten

16690



Trotten

16690

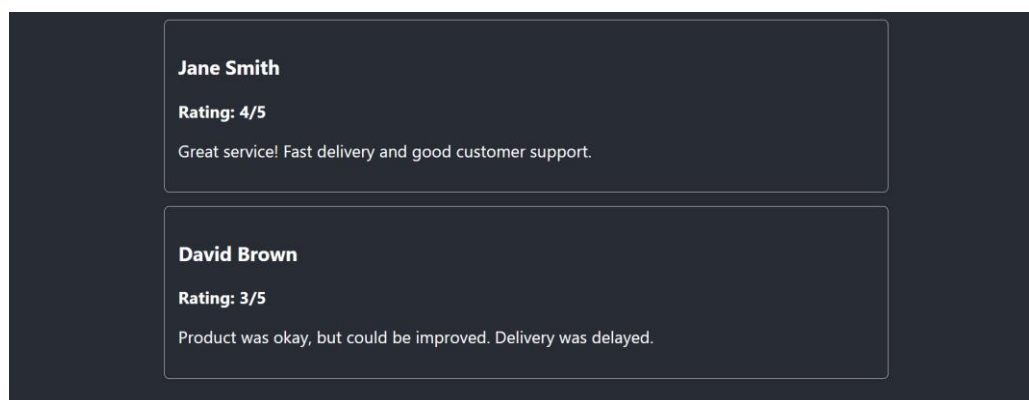
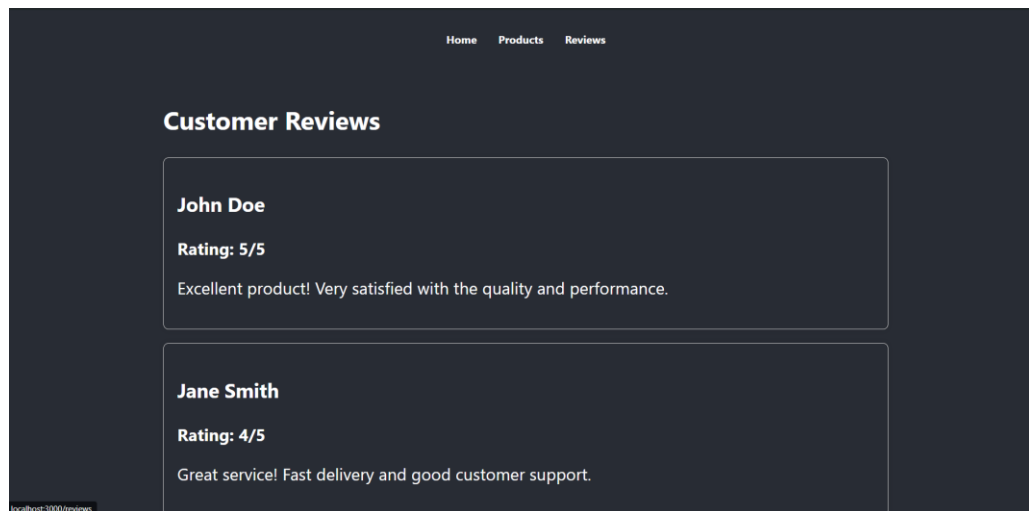




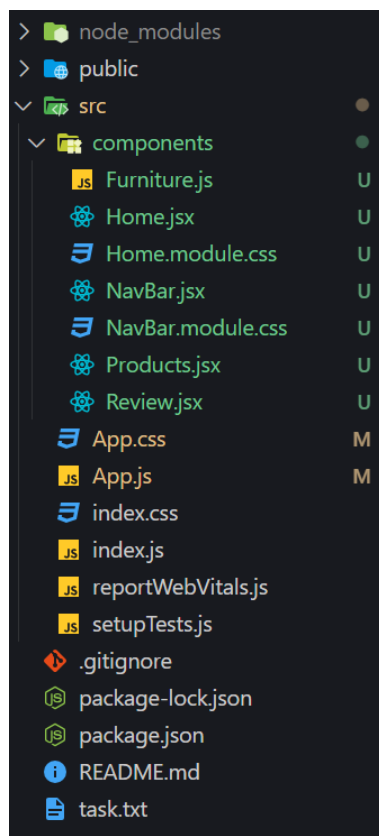


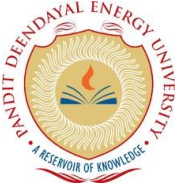
9 | Page

Customer Review page



Folder Structure



	Course Name: Advanced Web Technology	EXPERIMENT NO. 6
	Course Code: 20CP314P	Branch: CSE
	Faculty: Komal Singh	Semester: VI
Submitted by: Harsh Shah Roll no: 21BCP359		

Objective: Working with properties in React

1. In the commercial furniture web-application add products with information like image, price, color, manufacturer using react props.

App.js

```
import './App.css';
import Home from './Home';
import { BrowserRouter, Routes, Route } from 'react-router-dom';
import Task1and2 from './Task1and2/Task1and2';
import Task3and4 from './Task3and4/Task3and4';
function App() {
  return (
    <div>
      <BrowserRouter>
        <Routes>
          <Route path="/" element={ <Home /> } > </Route>
          <Route path="/student" element={ <Task1and2 /> } > </Route>
          <Route path="/furniture" element={ <Task3and4 /> } > </Route>
        </Routes>
      </BrowserRouter>
    </div>
  );
}
export default App;
```

Home.jsx

```
import './App.css';
import React from 'react'
import { Link } from 'react-router-dom';
const Home = () => {
  return (
    <div>
      <header className="App-header" style={{ width: '100vw' }}>
        <h2>Practical 6</h2>
        <Link to="/student" className="link">Go to Task 1 and 2</Link>
        <br />
        <Link to="/furniture" className="link">Go to Task 3 and 4</Link>
      </header>
    </div>
  )
}
export default Home;
```

Task3and4.js

```
import "../App.css";
import Furniture from "../Furniture";

function Task3and4() {
  return (
    <div className="grid-container App-header App">
      <Furniture
        itemName="Linanas"
        itemPrice="29990"
        itemImg="https://www.ikea.com/in/en/images/products/linanaes-3-seat-sofa-with-chaise-longue-vissle-
dark-grey__1013909_pe829463_s5.jpg?f=xxs"
        color="Gray"
        manufacturer="Ravi Furnitures"
      />

      <Furniture
        itemName="Nodeland"
        itemPrice="15990"
        itemImg="https://www.ikea.com/in/en/images/products/nodeland-wardrobe-with-3-doors-medium-
brown__0836934_pe778612_s5.jpg?f=xxs"
        color="Brown"
        manufacturer="New india Furnitures"
      />

      <Furniture
        itemName="Nolmyra"
        itemPrice="3990"
        itemImg="https://www.ikea.com/in/en/images/products/renberget-swivel-chair-bomstad-
black__1025981_pe834276_s5.jpg?f=xxs"
        color="Black"
        manufacturer="IKEA"
      />

      <Furniture
        itemName="Oxberg"
        itemPrice="3690"
        itemImg="https://www.ikea.com/in/en/images/products/billy-oxberg-bookcase-with-door-
white__1094849_pe863651_s5.jpg?f=xxs"
        color="White"
        manufacturer="he Interior Hub"
      />

      <Furniture
        itemName="Knarrevik"
        itemPrice="899"
        itemImg="https://www.ikea.com/in/en/images/products/knarrevik-bedside-table-
black__0858302_pe669481_s5.jpg?f=xxs"
        color="Black"
        manufacturer="Furniture Nest"
      />

      <Furniture
        itemName="Tufjord"
        itemPrice="42990"
        itemImg="https://www.ikea.com/in/en/images/products/knarrevik-bedside-table-
black__0858302_pe669481_s5.jpg?f=xxs"
```

```
        color="Black"
        manufacturer="IKEA"
    />
</div>
);
}
export default Task3and4;
```

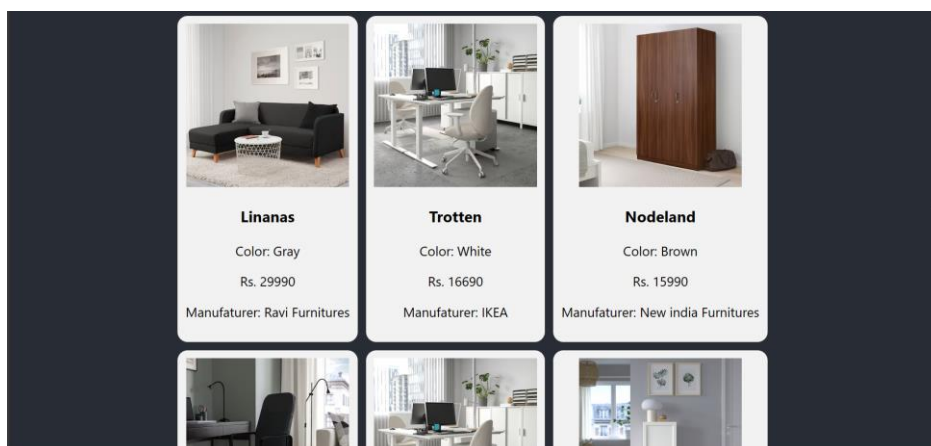
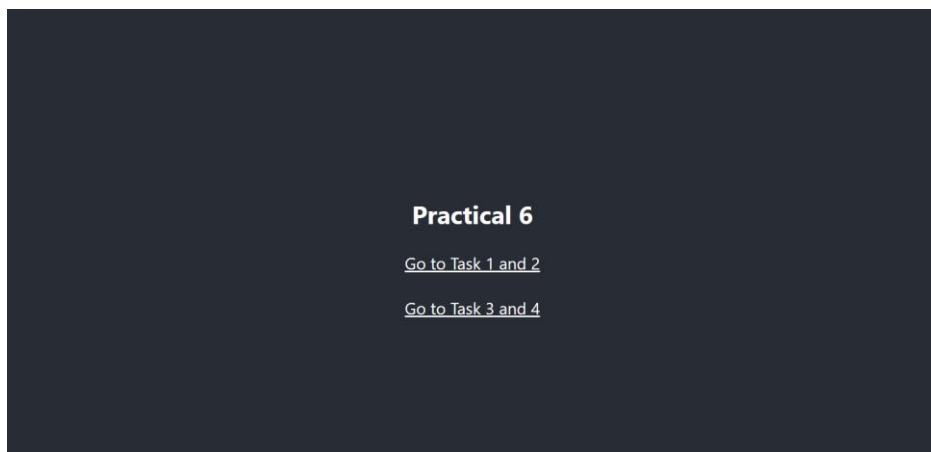
Furniture.js

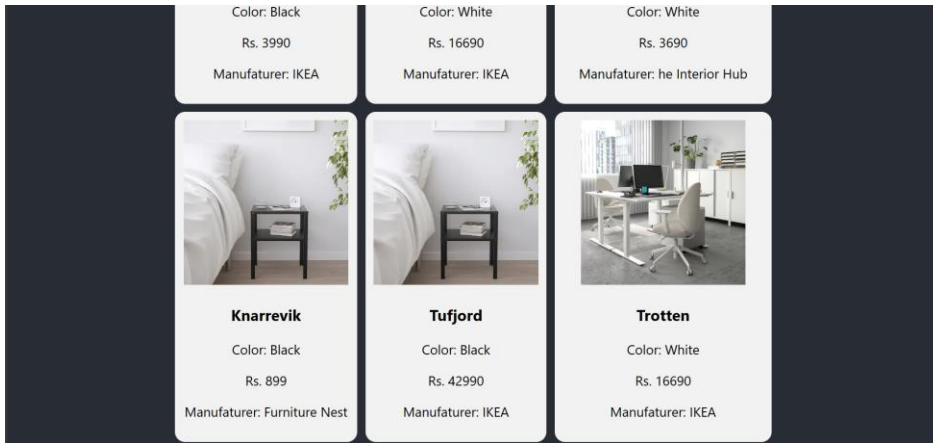
```
import "../App.css";

function Furniture(props) {
  return (
    <div className="grid-item">
      <img src={props.itemImg} alt={props.itemName} className="itemImg" />
      <h3 className="itemName">{props.itemName}</h3>
      <p className="itemPrice">Color: {props.color}</p>
      <p className="itemPrice">Rs. {props.itemPrice}</p>
      <p className="itemPrice">Manufaturer: {props.manufacturer}</p>
    </div>
  );
}
"IKEA",
};

export default Furniture;
```

Output





2. Add default values for all the furniture properties which are displayed in absence of corresponding information.

Modified Task3and4.js

```
import "../App.css";
import Furniture from "../Furniture";

function Task3and4() {
  return (
    <div className="grid-container App-header App">
      <Furniture
        itemName="Linanas"
        itemPrice="29990"
        itemImg="https://www.ikea.com/in/en/images/products/linanaes-3-seat-sofa-with-chaise-longue-vissle-dark-grey__1013909_pe829463_s5.jpg?f=xxs"
        color="Gray"
        manufacturer="Ravi Furnitures"
      />

      <Furniture />

      <Furniture
        itemName="Nodeland"
        itemPrice="15990"
        itemImg="https://www.ikea.com/in/en/images/products/nodeland-wardrobe-with-3-doors-medium-brown__0836934_pe778612_s5.jpg?f=xxs"
        color="Brown"
        manufacturer="New india Furnitures"
      />

      <Furniture
        itemName="Nolmyra"
        itemPrice="3990"
        itemImg="https://www.ikea.com/in/en/images/products/renberget-swivel-chair-bomstad-black__1025981_pe834276_s5.jpg?f=xxs"
        color="Black"
        manufacturer="IKEA"
      />

      <Furniture />
    </div>
  );
}
```

```

    <Furniture
      itemName="Oxberg"
      itemPrice="3690"
      itemImg="https://www.ikea.com/in/en/images/products/billy-oxberg-bookcase-with-door-
white__1094849_pe863651_s5.jpg?f=xxs"
      color="White"
      manufacturer="The Interior Hub"
    />

    <Furniture
      itemName="Knarrevik"
      itemPrice="899"
      itemImg="https://www.ikea.com/in/en/images/products/knarrevik-bedside-table-
black__0858302_pe669481_s5.jpg?f=xxs"
      color="Black"
      manufacturer="Furniture Nest"
    />

    <Furniture
      itemName="Tufjord"
      itemPrice="42990"
      itemImg="https://www.ikea.com/in/en/images/products/knarrevik-bedside-table-
black__0858302_pe669481_s5.jpg?f=xxs"
      color="Black"
      manufacturer="IKEA"
    />

    <Furniture />
  </div>
);
}
export default Task3and4;

```

Modified Furniture.js

```

import "../App.css";

function Furniture(props) {
  return (
    <div className="grid-item">
      <img src={props.itemImg} alt={props.itemName} className="itemImg" />
      <h3 className="itemName">{props.itemName}</h3>
      <p className="itemPrice">Color: {props.color}</p>
      <p className="itemPrice">Rs. {props.itemPrice}</p>
      <p className="itemPrice">Manufacturer: {props.manufacturer}</p>
    </div>
  );
}

Furniture.defaultProps = {
  itemName: "Trotten",
  itemPrice: "16690",
  itemImg:
    "https://www.ikea.com/in/en/images/products/trotten-desk-sit-stand-white__1040109_pe840587_s5.jpg?f=xxs",
  color: "White",
  manufacturer: "IKEA",
};
export default Furniture;

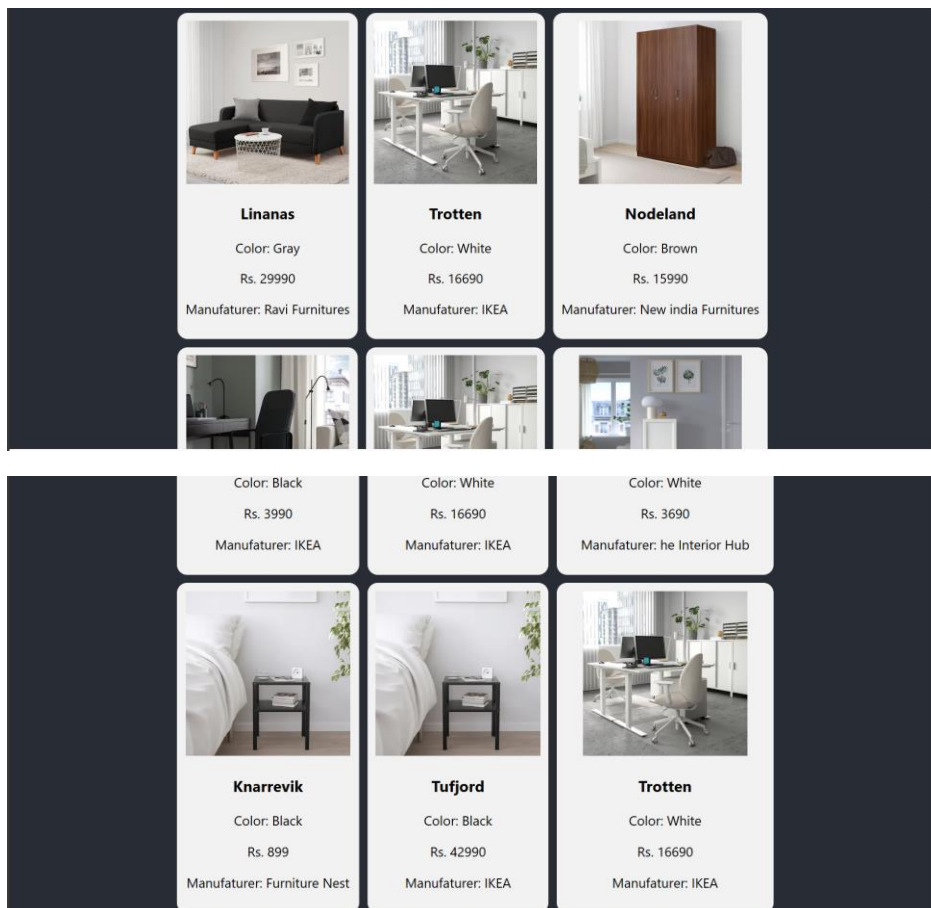
```

Output

Practical 6

[Go to Task 1 and 2](#)

[Go to Task 3 and 4](#)

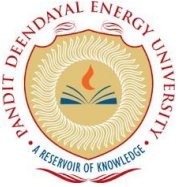


You can now view practical-6-1 in the browser.

Local: `http://localhost:3000`
On Your Network: `http://172.29.32.1:3000`

Note that the development build is not optimized.
To create a production build, use `npm run build`.

webpack compiled **successfully**

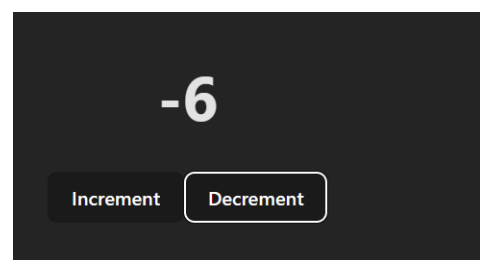
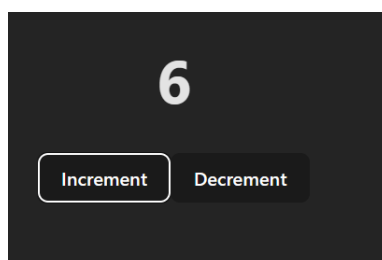
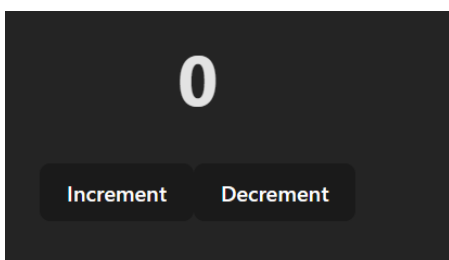
	Course Name: Advanced Web Technology	EXPERIMENT NO. 7
	Course Code: 20CP314P	Branch: CSE
	Faculty: Komal Singh	Semester: VI
Submitted by: Harsh Shah Roll no: 21BCP359		

Objective: Working with states in React

1. Create a counter based on react states using both the approaches class component as well as functional component.

Approach 1: Class Component

```
import React from "react";
class App extends React.Component {
  constructor(counter) {
    super(counter);
    this.state = {
      counter: 0,
    };
  }
  increment = () => {
    this.setState((prevState) => ({
      counter: prevState.counter + 1
    }));
  }
  decrement = () => {
    this.setState((prevState) => ({
      counter: prevState.counter - 1
    }));
  }
  render() {
    return (
      <div>
        <h1>{this.state.counter}</h1>
        <button onClick={this.increment}>Increment</button>
        <button onClick={this.decrement}>Decrement</button>
      </div>
    );
  }
}
export default App;
```

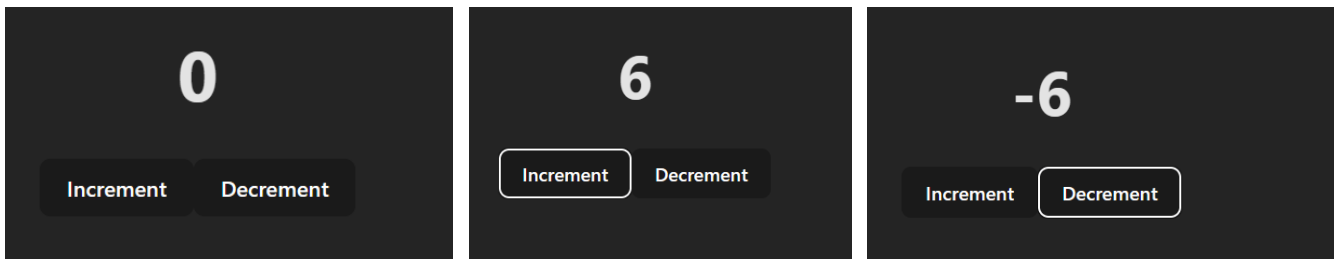


Approach 2: Functional Component

```
import { useState } from "react";
function App2() {
  const [count, setCount] = useState(0);

  const increment = () => {
    setCount(count + 1);
  };

  const decrement = () => {
    setCount(count - 1);
  };
  return (
    <div>
      <h1>{count}</h1>
      <button onClick={increment}>Increment</button>
      <button onClick={decrement}>Decrement</button>
    </div>
  );
}
export default App2;
```

**2. In the commercial furniture web-application add a like button under each product in the products page using react states.**

```
import Furniture from './Furniture';

function App3() {
  return (
    <div className="grid-container App-header App">
      <Furniture itemName="Linanas" itemPrice="29990"
itemImg="https://www.ikea.com/in/en/images/products/linanaes-3-seat-sofa-with-chaise-longue-vissle-dark-
grey__1013909_pe829463_s5.jpg?f=xxs" />

      <Furniture />

      <Furniture itemName="Nodeland" itemPrice="15990"
itemImg="https://www.ikea.com/in/en/images/products/nodeland-wardrobe-with-3-doors-medium-
brown__0836934_pe778612_s5.jpg?f=xxs" />

      <Furniture itemName="Nolmyra" itemPrice="3990"
itemImg="https://www.ikea.com/in/en/images/products/renberget-swivel-chair-bomstad-
black__1025981_pe834276_s5.jpg?f=xxs" />

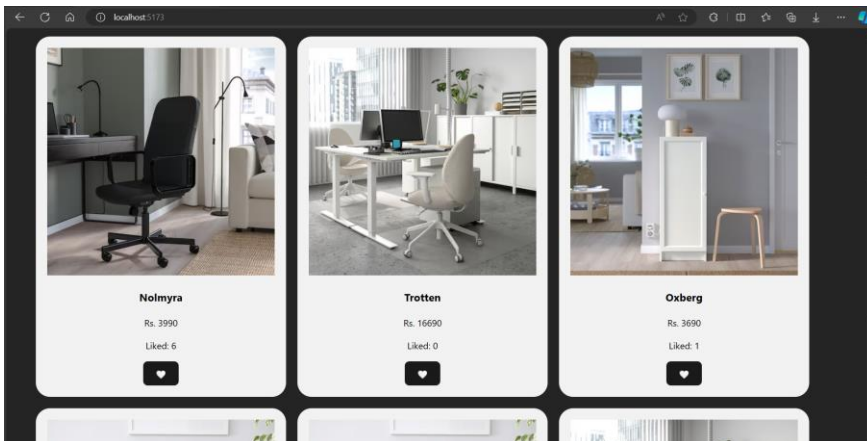
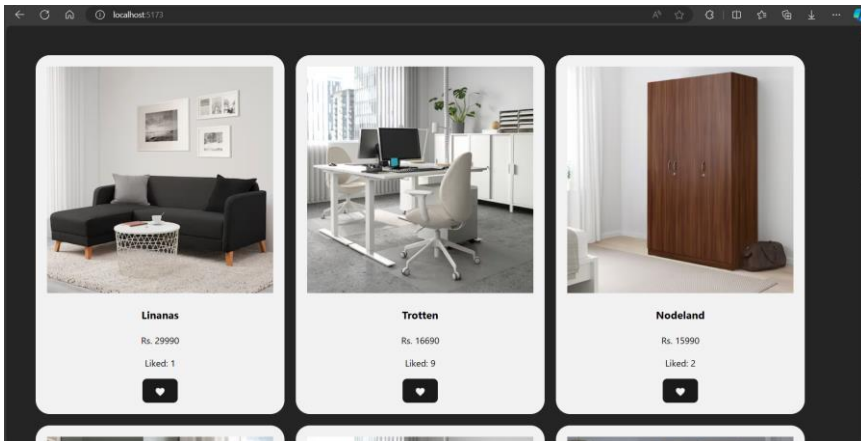
      <Furniture />
    </div>
  );
}
```

```
<Furniture itemName="Knarrevik" itemPrice="899"
itemImg="https://www.ikea.com/in/en/images/products/knarrevik-bedside-table-
black__0858302_pe669481_s5.jpg?f=xxs" />
```

```
<Furniture itemName="Tufjord" itemPrice="42990"
itemImg="https://www.ikea.com/in/en/images/products/knarrevik-bedside-table-
black__0858302_pe669481_s5.jpg?f=xxs" />
```

```
</div>
);
}
export default App3;
```

Output



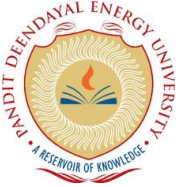
```
PowerShell 7.4.1
PS D:\DEV\Web Development\PDEU-AWT\Practical_7\practical-7-1> npm run dev

> practical-7-1@0.0.0 dev
> vite

Re-optimizing dependencies because vite config has changed

VITE v5.1.5 ready in 485 ms

➔ Local:   http://localhost:5173/
➔ Network: use --host to expose
➔ press h + enter to show help
11:44:41 am [vite] page reload src/main.jsx
11:44:51 am [vite] page reload src/main.jsx (x2)
11:50:59 am [vite] page reload src/main.jsx (x3)
11:51:37 am [vite] hmr update /src/Furniture.jsx
11:52:24 am [vite] hmr update /src/Furniture.jsx (x2)
11:52:27 am [vite] hmr update /src/Furniture.jsx (x3)
11:53:48 am [vite] hmr update /src/Furniture.jsx (x4)
11:54:00 am [vite] hmr update /src/Furniture.jsx (x5)
```

	Course Name: Advanced Web Technology	EXPERIMENT NO. 8
	Course Code: 20CP314P	Branch: CSE
	Faculty: Komal Singh	Semester: VI
Submitted by: Harsh Shah Roll no: 21BCP359		

Objective: Working with forms in React

1. Create a customer review form using react forms for the commercial furniture web-application with multiple fields such as name of the customer(text), the ratings for the customer experience out of 5(a drop-down menu), whether interested in buying again in future (check-box yes or no) and message from the customer(text-area) also ensure that the message should be less than 25 characters.

index.js

```
import React from "react";
import ReactDOM from "react-dom/client";
import App from "./App.jsx";
import "./index.css";
import { BrowserRouter } from "react-router-dom";

ReactDOM.createRoot(document.getElementById("root")).render(
  <React.StrictMode>
    <BrowserRouter>
      <App />
    </BrowserRouter>
  </React.StrictMode>
);
```

App.js

```
import { Route, Navigate, Routes } from "react-router-dom";
import CustomerReviewForm from "./CustomerReviewForm";
import ThankYouPage from "./ThankyouPage";

function App() {
  return (
    <div>
      <Routes>
        <Route path="/" element={ <CustomerReviewForm /> } />
        <Route path="/thankyou" element={ <ThankYouPage /> } />
        <Route path="*" element={ <Navigate to="/" /> } </Route>
      </Routes>
    </div>
  );
}

export default App;
```

CustomerReviewForm.jsx

```

import { useEffect, useState } from "react";
import { useNavigate } from "react-router-dom";

const CustomerReviewForm = () => {
  const navigate = useNavigate();
  const [submitted, setSubmitted] = useState(false);
  const [name, setName] = useState("");
  const [rating, setRating] = useState("");
  const [interested, setInterested] = useState(false);
  const [message, setMessage] = useState("");

  const submitForm = (e) => {
    e.preventDefault();
    setSubmitted(true);
  };

  useEffect(() => {
    if (submitted) {
      console.log("Form submitted: ", name, rating, interested, message);
      navigate("/thankyou", {
        replace: true,
        state: { name, rating, interested, message },
      });
    }
  }, [submitted, name, rating, interested, message, navigate]);

  return (
    <div>
      <h1>Commercial Furniture Web Application</h1>
      <form onSubmit={(e) => submitForm(e)}>
        <table>
          <tbody>
            <tr>
              <td>
                <label>Name:</label>
              </td>
              <td>
                <input
                  type="text"
                  value={name}
                  onChange={(e) => setName(e.target.value)}
                />
              </td>
            </tr>
            <tr>
              <td>
                <label>Rating:</label>
              </td>
              <td>
                <select
                  value={rating}
                  onChange={(e) => setRating(e.target.value)}
                >
                  <option value="">Select</option>
                  <option value="1">1</option>
                  <option value="2">2</option>
                </select>
              </td>
            </tr>
          </tbody>
        </table>
      </form>
    </div>
  );
};

```

```

        <option value="3">3</option>
        <option value="4">4</option>
        <option value="5">5</option>
      </select>
    </td>
  </tr>
  <tr>
    <td>
      <label>
        Interested in buying again in future:
      </label>
    </td>
    <td>
      <label className="checkbox-label">
        <input
          type="checkbox"
          checked={interested}
          onChange={(e) =>
            setInterested(e.target.checked)
          }
        />
        Yes
      </label>
    </td>
  </tr>
  <tr>
    <td>
      <label>Message (Max 25 characters):</label>
    </td>
    <td>
      <textarea
        value={message}
        onChange={(e) => setMessage(e.target.value)}
        maxLength={25}
      />
    </td>
  </tr>
  <tr>
    <td colspan="2">
      <button type="submit">Submit</button>
    </td>
  </tr>
</tbody>
</table>
</form>
</div>
);
};
export default CustomerReviewForm;

```

ThankyouPage.jsx

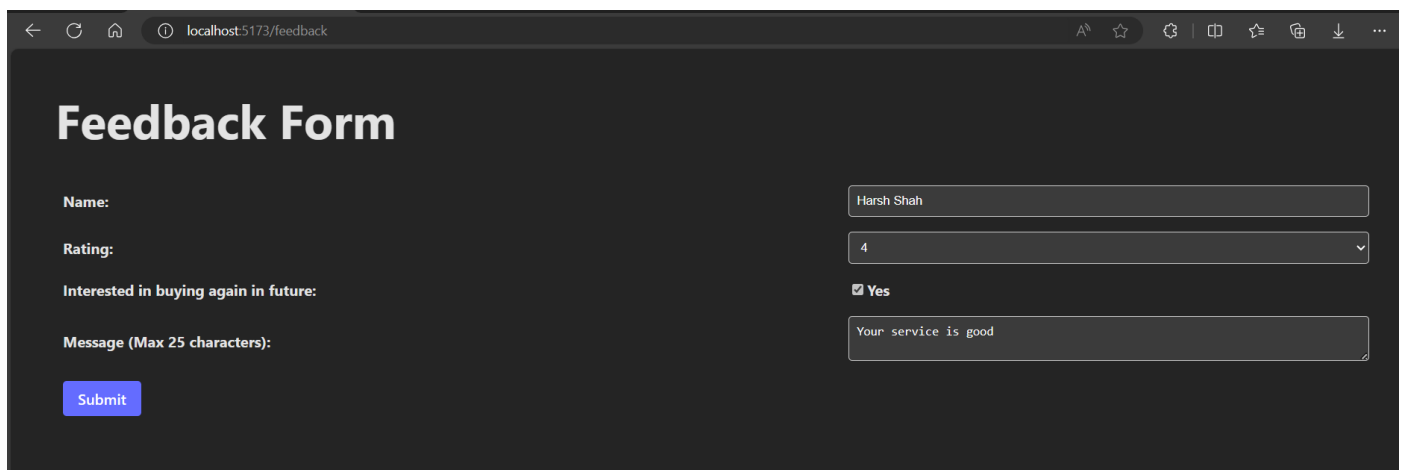
```
import { useLocation } from "react-router-dom";

const ThankYouPage = () => {
  const location = useLocation();
  const formData = location.state;

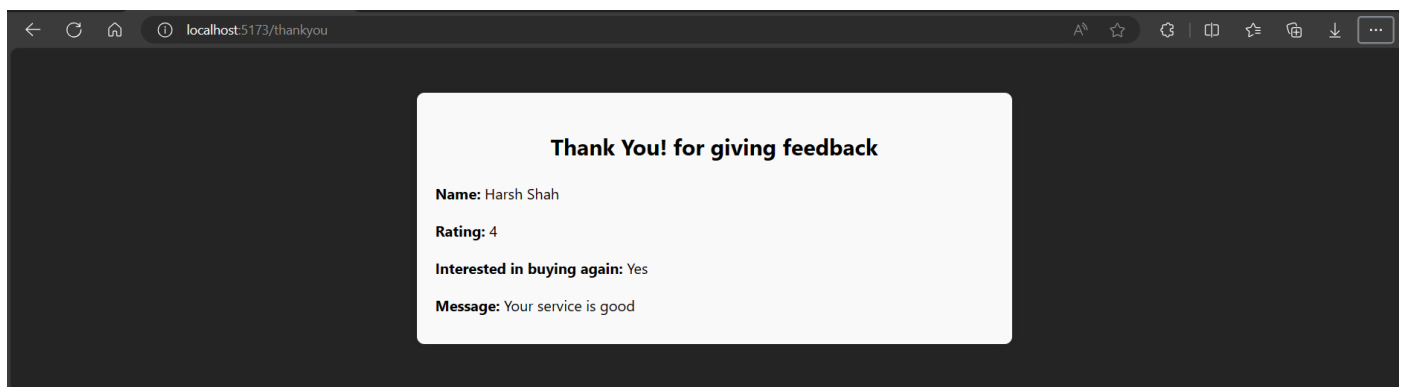
  return (
    <div>
      <h2>Thank You! for giving feedback</h2>
      <p>Name: {formData.name}</p>
      <p>Rating: {formData.rating}</p>
      <p>
        Interested in buying again: {formData.interested ? "Yes" : "No"}
      </p>
      <p>Message: {formData.message}</p>
    </div>
  );
};

export default ThankYouPage;
```

Output



A screenshot of a web browser showing a feedback form at localhost:5173/feedback. The form has a dark background with white text. It includes fields for Name (filled with 'Harsh Shah'), Rating (a dropdown menu showing '4'), Interested in buying again in future (a checkbox labeled 'Yes' which is checked), and Message (Max 25 characters, filled with 'Your service is good'). A blue 'Submit' button is at the bottom left.



A screenshot of a web browser showing a 'Thank You' confirmation page at localhost:5173/thankyou. The page has a dark background. A white box in the center contains the text 'Thank You! for giving feedback' and lists the submitted data: Name: Harsh Shah, Rating: 4, Interested in buying again: Yes, and Message: Your service is good.

2. Create a react form having fields like name (text), division (drop down menu) and message (text-area) using the useRef hook.

StudentData.jsx

```
import { useRef } from "react";
import { useNavigate } from "react-router-dom";
import "../FormData.css";

const StudentData = () => {
  const nameRef = useRef(null);
  const divisionRef = useRef(null);
  const messageRef = useRef(null);
  const navigate = useNavigate();

  const handleSubmit = (e) => {
    e.preventDefault();

    const name = nameRef.current.value;
    const division = divisionRef.current.value;
    const message = messageRef.current.value;

    navigate("/acknowledge", {
      state: {
        name,
        division,
        message,
      },
    });
  };

  return (
    <div>
      <h1>Student Details Form</h1>
      <form onSubmit={handleSubmit}>
        <table className="form-table">
          <tbody>
            <tr>
              <td>
                <label htmlFor="name">Name:</label>
              </td>
              <td>
                <input type="text" id="name" ref={nameRef} />
              </td>
            </tr>
            <tr>
              <td>
                <label htmlFor="division">Division:</label>
              </td>
              <td>
                <select id="division" ref={divisionRef}>
                  <option value="1">Division 1</option>
                  <option value="2">Division 2</option>
                  <option value="3">Division 3</option>
                  <option value="4">Division 4</option>
                  <option value="5">Division 5</option>
                  <option value="6">Division 6</option>
                </select>
              </td>
            </tr>
          </tbody>
        </table>
      </form>
    </div>
  );
};
```

```

    </tr>
    <tr>
      <td>
        <label htmlFor="message">Message:</label>
      </td>
      <td>
        <textarea
          id="message"
          ref={messageRef}
        ></textarea>
      </td>
    </tr>
    <tr>
      <td colSpan="2">
        <button type="submit">Submit</button>
      </td>
    </tr>
  </tbody>
</table>
</form>
</div>
);
};

```

```
export default StudentData;
```

Acknowledge.jsx

```

import { useLocation } from "react-router-dom";
import "../ThankYouPage.css";

const Acknowledge = () => {
  const location = useLocation();
  const formData = location.state;

  return (
    <div className="thank-you-container">
      <h2 className="thank-you-heading">Thank You for Your Feedback!</h2>
      <div className="thank-you-details">
        <p>
          <strong>Name:</strong> {formData.name}
        </p>
        <p>
          <strong>Division:</strong> {formData.division}
        </p>
        <p>
          <strong>Message:</strong> {formData.message}
        </p>
      </div>
    </div>
  );
};
export default Acknowledge;

```

FormData.css

```
.form-table {  
  width: 100%;  
  border-collapse: collapse;  
}  
  
.form-table td {  
  padding: 8px;  
}  
  
.form-table label {  
  font-weight: bold;  
}  
  
.form-table input[type="text"],  
.form-table select,  
.form-table textarea {  
  width: 100%;  
  padding: 8px;  
  border: 1px solid #ccc;  
  border-radius: 4px;  
  box-sizing: border-box;  
}  
  
.form-table button {  
  padding: 8px 16px;  
  border: none;  
  border-radius: 4px;  
  background-color: #646cff;  
  color: #fff;  
  cursor: pointer;  
}  
  
.form-table button:hover {  
  background-color: #535bf2;  
}
```

ThankYouPage.css

```
.thank-you-container {  
  max-width: 600px;  
  margin: 0 auto;  
  padding: 20px;  
  background-color: #f9f9f9;  
  border-radius: 8px;  
  box-shadow: 0 0 10px rgba(0, 0, 0, 0.1);  
  color: black;  
}  
  
.thank-you-heading {  
  text-align: center;  
  color: black;  
}  
  
.thank-you-details {  
  margin-top: 20px;
```

20CP314P

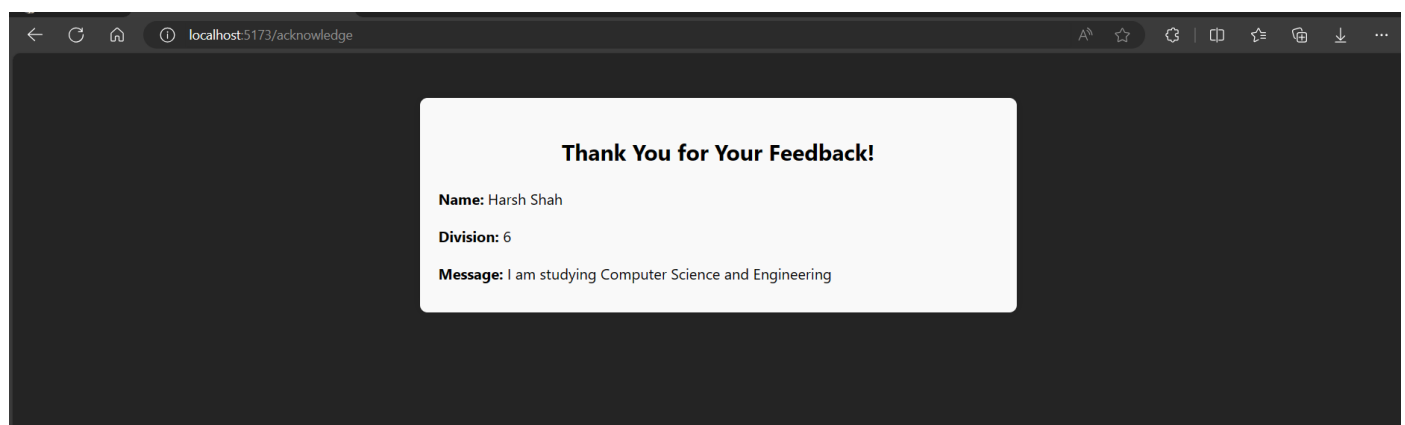
21BCP359

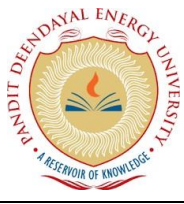
}

```
.thank-you-details p {  
  margin-bottom: 10px;  
}
```

```
.thank-you-details p strong {  
  font-weight: bold;  
}
```

Output



	Course Name: Advanced Web Technology	EXPERIMENT NO. 9
	Course Code: 20CP314P	Branch: CSE
	Faculty: Komal Singh	Semester: VI
Submitted by: Harsh Shah Roll no: 21BCP359		

Objective: Working with React hooks

1. Write react programs displaying the usage of each of the below given react hooks:

a. `useReducer`

UseReducer.jsx

```
import { useReducer } from "react";
import { ACTION_TYPES } from "../postActionTypes";
import { INITIAL_STATE, postReducer } from "../postReducer";

const Post = () => {
  const [state, dispatch] = useReducer(postReducer, INITIAL_STATE);

  const handleFetch = () => {
    dispatch({ type: ACTION_TYPES.FETCH_START });
    fetch("https://randomuser.me/api/")
      .then((res) => {
        return res.json();
      })
      .then((data) => {
        const firstName = data.results[0].name.first;
        const lastName = data.results[0].name.last;

        const greet = { message: "Hi " + firstName + " " + lastName };

        dispatch({ type: ACTION_TYPES.FETCH_SUCCESS, payload: greet });
      })
      .catch(() => {
        dispatch({ type: ACTION_TYPES.FETCH_ERROR });
      });
  };

  return (
    <div>
      <h1>useReducer</h1>
      <button onClick={handleFetch}>
        {state.loading ? "Wait..." : "Fetch the post"}
      </button>
      <p>{state.post?.message}</p>
      <span>{state.error && "Something went wrong!"}</span>
    </div>
  );
};

export default Post;
```

postreducer.js

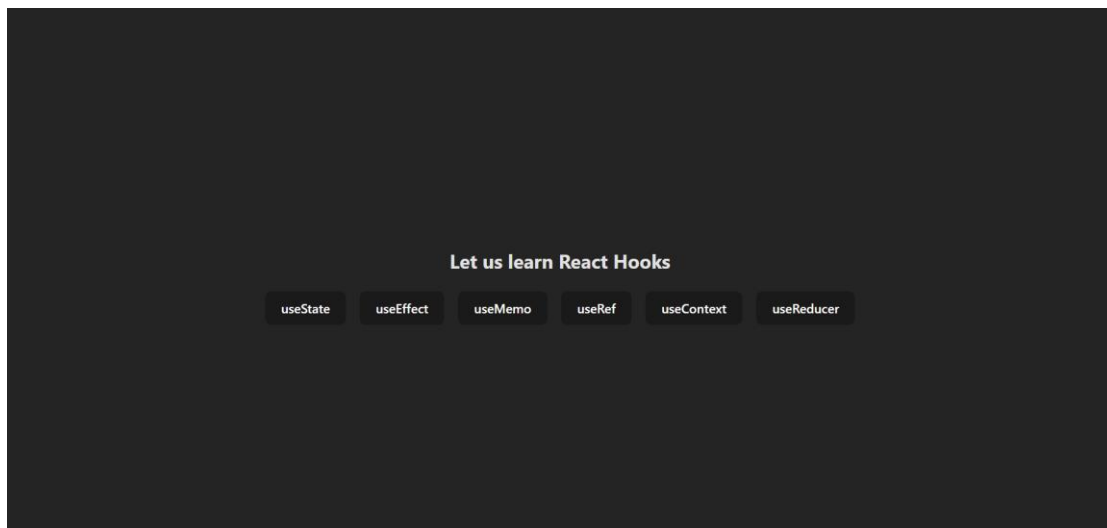
```
import { ACTION_TYPES } from "../postActionTypes";

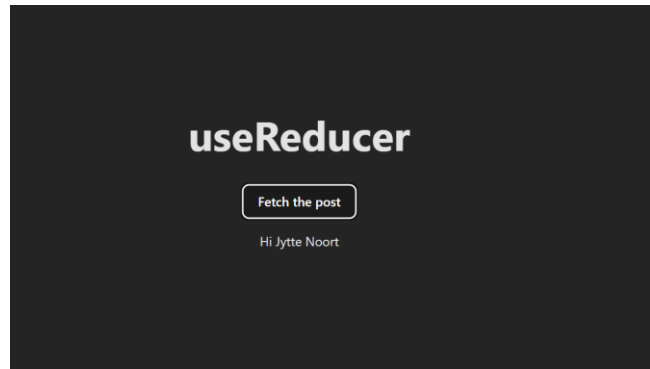
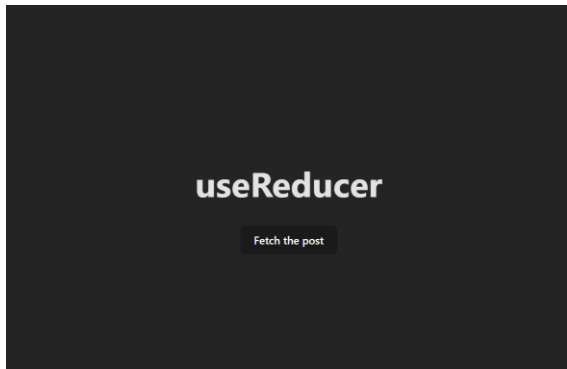
export const INITIAL_STATE = {
  loading: false,
  post: {},
  error: false,
};

export const postReducer = (state, action) => {
  switch (action.type) {
    case ACTION_TYPES.FETCH_START:
      return {
        loading: true,
        error: false,
        post: {},
      };
    case ACTION_TYPES.FETCH_SUCCESS:
      return {
        ...state,
        loading: false,
        post: action.payload,
      };
    case ACTION_TYPES.FETCH_ERROR:
      return {
        error: true,
        loading: false,
        post: {},
      };
    default:
      return state;
  }
};
```

postActionTypes.js

```
export const ACTION_TYPES = {
  FETCH_START: "FETCH_START",
  FETCH_SUCCESS: "FETCH_SUCCESS",
  FETCH_ERROR: "FETCH_ERROR",
};
```





b. useMemo

```
import { useState, useMemo } from "react";
import "../App.css";
function UseMemo() {
  const [count, setCount] = useState(0);
  const [darkTheme, setDarkTheme] = useState(false);
  const squareResult = useMemo(() => {
    delayFunction(count);
  }, [count]);
  const themeStyles = useMemo(() => {
    return {
      backgroundColor: darkTheme ? "blue" : "red",
      color: darkTheme ? "red" : "blue",
    };
  }, [darkTheme]);

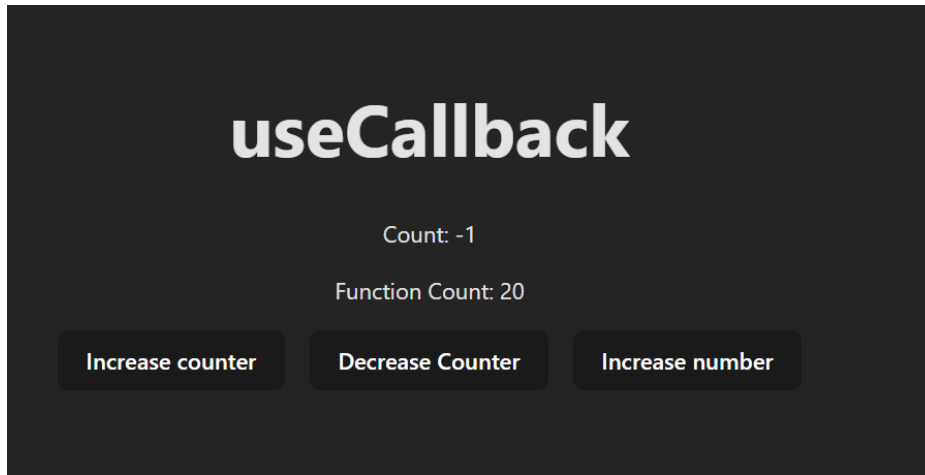
  return (
    <>
      <div className="card">
        <h1>useMemo</h1>
        <label>
          Number: { " "}
          <input
            type="number"
            value={count}
            onChange={(e) => setCount(parseInt(e.target.value))}
          />
        </label>

        <p>Square: {squareResult}</p>

        <button onClick={() => setDarkTheme((prevTheme) => !prevTheme)}>
          Change Theme
        </button>
        <p style={themeStyles}>Theme div</p>
      </div>
    </>
  );
}

function delayFunction(n) {
  for (let i = 0; i < 1000000000; i++);
  return n * n;
}
export default UseMemo;
```





d. useContext

```
import { useState, createContext, useContext } from "react";
```

```
const UserContext = createContext();
```

```
function useContext() {
  const [user] = useState("AWT Student");
```

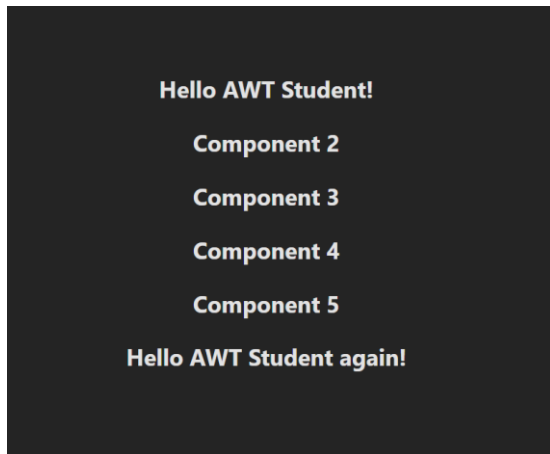
```
  return (
    <UserContext.Provider value={user}>
      <h2>{`Hello ${user}!`} </h2>
      <Component2 />
    </UserContext.Provider>
  );
}
```

```
function Component2() {
  return (
    <>
      <h2>Component 2</h2>
      <Component3 />
    </>
  );
}
```

```
function Component3() {
  return (
    <>
      <h2>Component 3</h2>
      <Component4 />
    </>
  );
}
```

```
function Component4() {
  return (
    <>
      <h2>Component 4</h2>
      <Component5 />
    </>
  );
}
```

```
function Component5() {
  const user = useContext(UserContext);
  return (
    <>
      <h2>Component 5</h2>
      <h2>`Hello ${user} again!`</h2>
    </>
  );
}
export default UseContext;
```



e. useEffect

```
import { useEffect, useState } from "react";
import "../App.css";
function UseEffect() {
  const [count, setCount] = useState(0);
  const [numberProperty, setNumberProperty] = useState("");

  useEffect(() => {
    if (count % 2 === 0) {
      setNumberProperty(`Even Number: ${count}`);
    } else {
      setNumberProperty(`Odd Number: ${count}`);
    }
    console.log("useEffect changed");
  }, [count]);

  return (
    <>
      <div className="card">
        <h1>useEffect</h1>
        <button onClick={() => setCount((count) => count + 1)}>
          Click Me!
        </button>
        <p>{numberProperty}</p>
      </div>
    </>
  );
}
export default UseEffect;
```



2. Design a custom hook that helps you declare a default value of a component

Customhook.jsx

```
import useFetch from "./useFetch";
const Customhook = () => {
  const data = useFetch("https://jsonplaceholder.typicode.com/todos");
  return (
    <div className="card">
      <h1>Custom Hook</h1>
      {data &&
        data.map((item) => {
          return <p key={item.id}>{item.title}</p>;
        })
      }
    </div>
  );
};
export default Customhook;
```

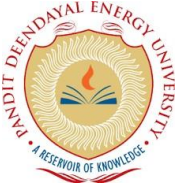
useFetch.js

```
import { useState, useEffect } from "react";
const useFetch = (url) => {
  const [data, setData] = useState(null);

  useEffect(() => {
    fetch(url)
      .then((res) => res.json())
      .then((data) => setData(data));
  }, [url]);

  return data;
};
export default useFetch;
```



	Course Name: Advanced Web Technology	EXPERIMENT NO. 10
	Course Code: 20CP314P	Branch: CSE
	Faculty: Komal Singh	Semester: VI
Submitted by: Harsh Shah Roll no: 21BCP359		

Objective: Implement a web-app using Django framework

1. Design a 'Hello World' Web-application using Django framework *views.py*

```
from django.shortcuts import render
```

```
# Create your views here.
```

```
from django.http import HttpResponse
```

```
def hello_world(request):
    return HttpResponse("Hello World!")
```

urls.py

```
"""
```

URL configuration for Lab_10 project.

The `urlpatterns` list routes URLs to views. For more information please see:

<https://docs.djangoproject.com/en/4.2/topics/http/urls/>

Examples:

Function views

1. Add an import: from my_app import views
2. Add a URL to urlpatterns: path("", views.home, name='home')

Class-based views

1. Add an import: from other_app.views import Home
2. Add a URL to urlpatterns: path("", Home.as_view(), name='home')

Including another URLconf

1. Import the include() function: from django.urls import include, path
2. Add a URL to urlpatterns: path('blog/', include('blog.urls'))

```
"""
```

```
from django.urls import path
```

```
from hello_world_app import views
```

```
urlpatterns = [
    path('hello/', views.hello_world, name='hello_world'),
]
```

```
PS D:\SEM-6\Advanced Web Technology> pip install django
Requirement already satisfied: django in c:\python39\lib\site-packages (4.2.11)
Requirement already satisfied: asgiref<4,>=3.6.0 in c:\python39\lib\site-packages (from django) (3.8.1)
Requirement already satisfied: sqlparse>=0.3.1 in c:\python39\lib\site-packages (from django) (0.5.0)
Requirement already satisfied: tzdata in c:\python39\lib\site-packages (from django) (2024.1)
Requirement already satisfied: typing-extensions>=4 in c:\python39\lib\site-packages (from asgiref<4,>=3.6.0->django) (4.7.1)

[notice] A new release of pip is available: 23.3.1 -> 24.0
[notice] To update, run: python.exe -m pip install --upgrade pip
```

```
PS D:\SEM-6\Advanced Web Technology> django-admin startproject Lab_10
PS D:\SEM-6\Advanced Web Technology> cd Lab_10
PS D:\SEM-6\Advanced Web Technology\Lab_10> python manage.py startapp hello_world_app
PS D:\SEM-6\Advanced Web Technology\Lab_10> python manage.py runserver
Watching for file changes with StatReloader
Performing system checks...

System check identified no issues (0 silenced).

You have 18 unapplied migration(s). Your project may not work properly until you apply the migrations for app(s): admin, auth, contenttypes, sessions.
Run 'python manage.py migrate' to apply them.
April 25, 2024 - 13:14:38
Django version 4.2.11, using settings 'Lab_10.settings'
Starting development server at http://127.0.0.1:8000/
Quit the server with CTRL-BREAK.

[25/Apr/2024 13:14:45] "GET / HTTP/1.1" 200 10664
Not Found: /favicon.ico
[25/Apr/2024 13:14:50] "GET /favicon.ico HTTP/1.1" 404 2110
[25/Apr/2024 13:15:16] "GET / HTTP/1.1" 200 10664
Not Found: /hello/
[25/Apr/2024 13:15:29] "GET /hello/ HTTP/1.1" 404 2095
Not Found: /hello
[25/Apr/2024 13:15:35] "GET /hello HTTP/1.1" 404 2092
Not Found: /hello/
[25/Apr/2024 13:15:47] "GET /hello/ HTTP/1.1" 404 2095
Not Found: /hello/
[25/Apr/2024 13:16:05] "GET /hello/ HTTP/1.1" 404 2095
PS D:\SEM-6\Advanced Web Technology\Lab_10> pip install django
Requirement already satisfied: django in c:\python39\lib\site-packages (4.2.11)
Requirement already satisfied: asgiref<4,>=3.6.0 in c:\python39\lib\site-packages (from django) (3.8.1)
Requirement already satisfied: sqlparse>=0.3.1 in c:\python39\lib\site-packages (from django) (0.5.0)
Requirement already satisfied: tzdata in c:\python39\lib\site-packages (from django) (2024.1)
Requirement already satisfied: typing-extensions>=4 in c:\python39\lib\site-packages (from asgiref<4,>=3.6.0->django) (4.7.1)
```

```
PS D:\SEM-6\Advanced Web Technology\Lab_10> python manage.py runserver
Watching for file changes with StatReloader
Performing system checks...

System check identified no issues (0 silenced).

You have 18 unapplied migration(s). Your project may not work properly until you apply the migrations for app(s): admin, auth, contenttypes, sessions.
Run 'python manage.py migrate' to apply them.
April 25, 2024 - 13:17:37
Django version 4.2.11, using settings 'Lab_10.settings'
Starting development server at http://127.0.0.1:8000/
Quit the server with CTRL-BREAK.

[25/Apr/2024 13:17:43] "GET / HTTP/1.1" 200 10664
Not Found: /hello/
[25/Apr/2024 13:17:54] "GET /hello/ HTTP/1.1" 404 2095
D:\SEM-6\Advanced Web Technology\Lab_10\Lab_10\urls.py changed, reloading.
Watching for file changes with StatReloader
Performing system checks...

System check identified no issues (0 silenced).

You have 18 unapplied migration(s). Your project may not work properly until you apply the migrations for app(s): admin, auth, contenttypes, sessions.
Run 'python manage.py migrate' to apply them.
April 25, 2024 - 13:20:55
Django version 4.2.11, using settings 'Lab_10.settings'
Starting development server at http://127.0.0.1:8000/
Quit the server with CTRL-BREAK.

Not Found: /
[25/Apr/2024 13:20:58] "GET / HTTP/1.1" 404 2079
[25/Apr/2024 13:21:08] "GET /hello/ HTTP/1.1" 200 12
```



127.0.0.1:8000/hello/



127.0.0.1:8000/hello/

Hello World!

2. Perform CRUD commands in a designed model using the default SQLite database extension in Django.

models.py

```
from django.db import models
```

```
class MyModel(models.Model):
```

```
    name = models.CharField(max_length=100)
```

```
    description = models.TextField()
```

```
    class Meta:
```

```
        app_label = 'hello_world_app'
```

settings.py

```
INSTALLED_APPS = [  
    'django.contrib.admin',  
    'django.contrib.auth',  
    'django.contrib.contenttypes',  
    'django.contrib.sessions',  
    'django.contrib.messages',  
    'django.contrib.staticfiles',  
    'hello_world_app'  
]
```

```
PS D:\SEM-6\Advanced Web Technology\Lab_10> python manage.py makemigrations
```

```
No changes detected
```

```
PS D:\SEM-6\Advanced Web Technology\Lab_10> python manage.py migrate
```

```
Operations to perform:
```

```
  Apply all migrations: admin, auth, contenttypes, sessions
```

```
Running migrations:
```

```
  No migrations to apply.
```

```
PS D:\SEM-6\Advanced Web Technology\Lab_10> python manage.py makemigrations hello_world_app
```

```
Migrations for 'hello_world_app':
```

```
  hello_world_app\migrations\0002_remove_mymodel_description.py
```

```
PS D:\SEM-6\Advanced Web Technology\Lab_10> python manage.py migrate
```

```
Operations to perform:
```

```
  Apply all migrations: admin, auth, contenttypes, hello_world_app, sessions
```

```
Running migrations:
```

```
  Applying hello_world_app.0002_remove_mymodel_description... OK
```

Output:

View SQL

```
PS D:\SEM-6\Advanced Web Technology\Lab_10> py manage.py sqlmigrate hello_world_a
pp 0001
BEGIN;
--
-- Create model MyModel
--
CREATE TABLE "hello_world_app_mymodel" ("id" integer NOT NULL PRIMARY KEY AUTOINC
REMENT, "name" varchar(100) NOT NULL, "description" text NOT NULL);
COMMIT;
```

Add Records

```
PS D:\SEM-6\Advanced Web Technology\Lab_10> python manage.py shell
Python 3.9.6 (tags/v3.9.6:db3ff76, Jun 28 2021, 15:26:21) [MSC v.1929 64 bit (AMD
64)]
Type 'copyright', 'credits' or 'license' for more information
IPython 8.17.2 -- An enhanced Interactive Python. Type '?' for help.
```

```
In [1]:
```

```
In [1]: from hello_world_app.models import MyModel

In [2]: MyModel.objects.all()
Out[2]: <QuerySet []>

In [3]: text = hello_world_app(name="Adit Shah")
-----
NameError                                Traceback (most recent call last)
Cell In[3], line 1
----> 1 text = hello_world_app(name="Adit Shah")

NameError: name 'hello_world_app' is not defined

In [4]: text = MyModel(name="Adit Shah")

In [5]: text.save()

In [6]: MyModel.objects.all().values()
Out[6]: <QuerySet [{'id': 1, 'name': 'Adit Shah'}]>
```

Update Records

```
In [9]: x = MyModel.objects.all()[0]

In [10]: x.name
Out[10]: 'Adit Shah'

In [11]: x.name = "21BCP419"

In [12]: x.save()

In [13]: MyModel.objects.all().values()
Out[13]: <QuerySet [{'id': 1, 'name': '21BCP419'}]>
```

Delete Records

```
In [15]: x = MyModel.objects.all()[0]

In [16]: x.name
Out[16]: '21BCP419'

In [17]: '21BCP419'
Out[17]: '21BCP419'

In [18]: x.delete()
Out[18]: (1, {'hello_world_app.MyModel': 1})

In [19]: MyModel.objects.all().values()
Out[19]: <QuerySet []>
```

3. Connect your Django web-application to PostgreSQL database and perform CRUD commands.

settings.py

```
DATABASES = {  
    'default': {  
        'ENGINE': 'django.db.backends.postgresql',  
        'NAME': 'user',  
        'USER': 'postgres',  
        'PASSWORD': 'password',  
        'HOST': 'localhost',  
        'PORT': '5432',  
    }  
}
```

Views.py

```
from django.shortcuts import render, redirect
```

```
from .models import Item
```

```
def index(request):
```

```
    items = Item.objects.all()
```

```
    return render(request, 'index.html', {'items': items})
```

```
def create(request):
```

```
    if request.method == 'POST':
```

```
        name = request.POST.get('name')
```

```
        description = request.POST.get('description')
```

```
        Item.objects.create(name=name, description=description)
```

```
        return redirect('index')
```

```
    return render(request, 'create.html')
```

```
def update(request, item_id):
```

```
    item = Item.objects.get(id=item_id)
```

```
    if request.method == 'POST':
```

```
        item.name = request.POST.get('name')
```

```
        item.description = request.POST.get('description')
```

```
        item.save()
```

```
        return redirect('index')

    return render(request, 'update.html', {'item': item})
```

```
def delete(request, item_id):
    item = Item.objects.get(id=item_id)
    item.delete()
    return redirect('index')
```

urls.py

```
from django.urls import path
from . import views

urlpatterns = [
    path("", views.index, name='index'),
    path('create/', views.create, name='create'),
    path('update/<int:item_id>/', views.update, name='update'),
    path('delete/<int:item_id>/', views.delete, name='delete'),
]
```

Create.html

```
<h1>Create Item</h1>
<form method="post">
    {% csrf_token %}
    <label for="name">Name:</label>
    <input type="text" name="name" required>
    <br>
    <label for="description">Description:</label>
    <textarea name="description" required></textarea>
    <br>
    <input type="submit" value="Create">
</form>
```

Index.html

```
<h1>Items</h1>

<ul>

    {% for item in items %}

        <li>{{ item.name }} - {{ item.description }}</li>

    {% endfor %}

</ul>

<a href="{% url 'create' %}">Create Item</a>
```

Update.html

```
<h1>Update Item</h1>

<form method="post">

    {% csrf_token %}

    <label for="name">Name:</label>

    <input type="text" name="name" value="{{ item.name }}" required>

    <br>

    <label for="description">Description:</label>

    <textarea name="description" required>{{ item.description }}</textarea>

    <br>

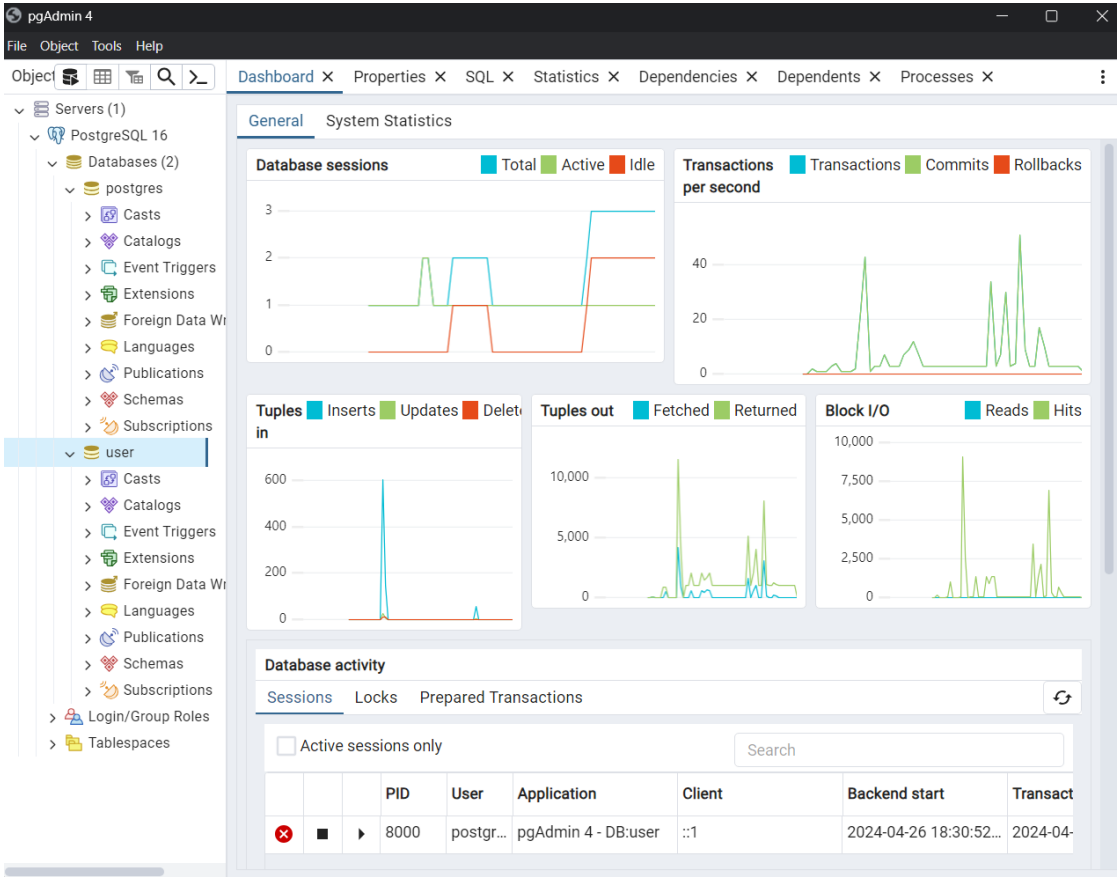
    <input type="submit" value="Update">

</form>
```

Output

```
PS D:\SEM-6\Advanced Web Technology\django\crud\crud_project> python manage.py makemigrations
Migrations for 'crud_app':
  crud_app\migrations\0001_initial.py
    - Create model Item
PS D:\SEM-6\Advanced Web Technology\django\crud\crud_project> python manage.py migrate
Operations to perform:
  Apply all migrations: admin, auth, contenttypes, crud_app, sessions
Running migrations:
  Applying crud_app.0001_initial... OK
PS D:\SEM-6\Advanced Web Technology\django\crud\crud_project> python manage.py runserver
Watching for file changes with StatReloader
Performing system checks...

System check identified no issues (0 silenced).
April 26, 2024 - 19:02:22
Django version 4.2.11, using settings 'crud_project.settings'
Starting development server at http://127.0.0.1:8000/
Quit the server with CTRL-BREAK.
```



Output:

127.0.0.1:8000

127.0.0.1:8000

Items

[Create Item](#)

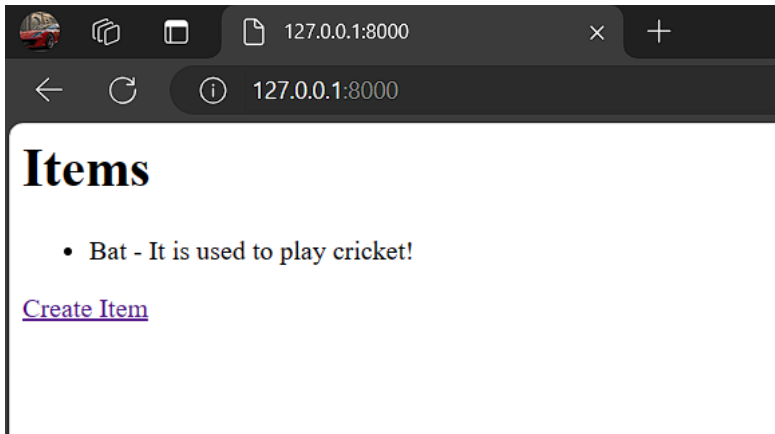
Add Records

Create Item

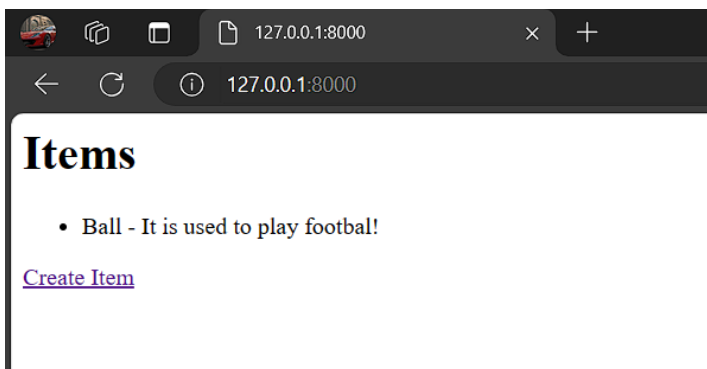
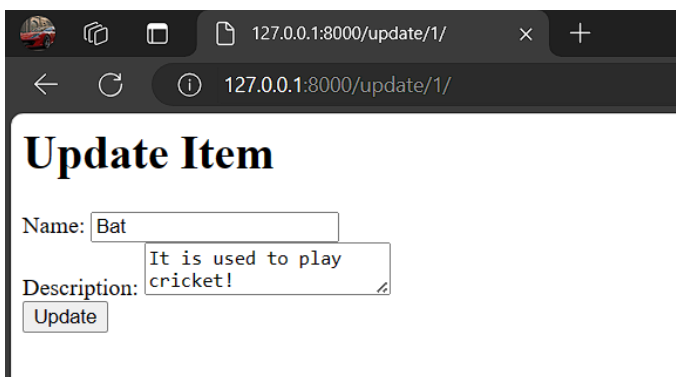
Name:

Description:

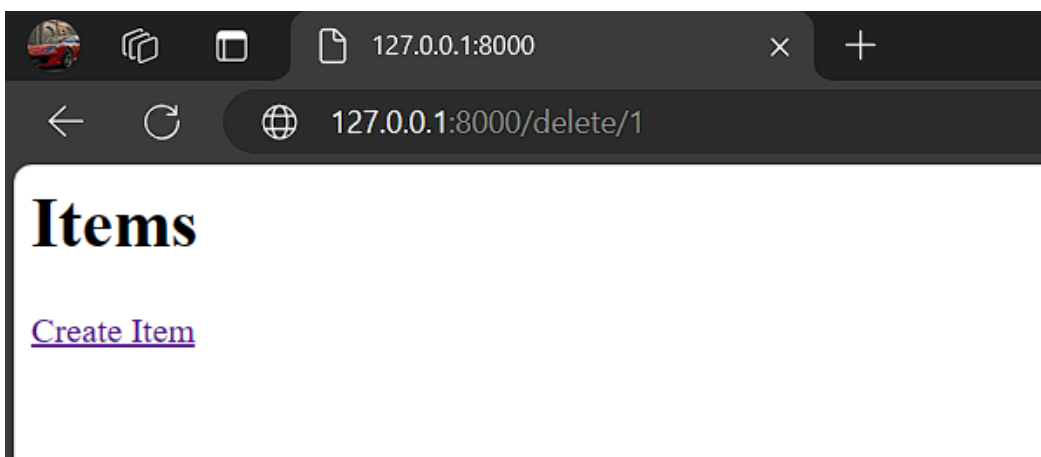


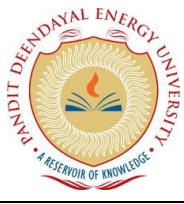


Update Records



Delete Records



	Course Name: Advanced Web Technology	EXPERIMENT NO. 11
	Course Code: 20CP314P	Branch: CSE
	Faculty: Komal Singh	Semester: VI
Submitted by: Harsh Shah Roll no: 21BCP359		

Objective: Implement a web-app using Flask framework

1. Design a 'Hello World' Web-application using Flask framework

```
from flask import Flask
```

```
app = Flask(__name__)
```

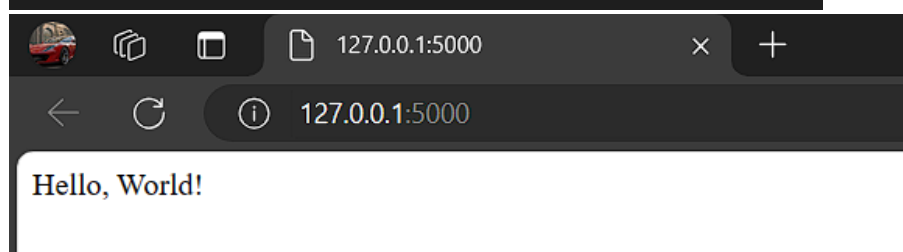
```
@app.route('/')  
def hello_world():
```

```
    return 'Hello, World!'
```

```
if __name__ == '__main__':
```

```
    app.run(debug=True)
```

```
PROBLEMS 7 TERMINAL OUTPUT PORTS DEBUG CONSOLE  
PS D:\SEM-6\Advanced Web Technology\Lab_11> pip install flask  
Requirement already satisfied: flask in c:\python39\lib\site-packages (2.0.1)  
Requirement already satisfied: Werkzeug>=2.0 in c:\python39\lib\site-packages (from flask) (2.0.1)  
Requirement already satisfied: Jinja2>=3.0 in c:\python39\lib\site-packages (from flask) (3.0.1)  
Requirement already satisfied: itsdangerous>=2.0 in c:\python39\lib\site-packages (from flask) (2.0.1)  
Requirement already satisfied: click>=7.1.2 in c:\python39\lib\site-packages (from flask) (8.0.1)  
Requirement already satisfied: colorama in c:\python39\lib\site-packages (from click>=7.1.2->flask) (0.4.4)  
Requirement already satisfied: MarkupSafe>=2.0 in c:\python39\lib\site-packages (from Jinja2>=3.0->flask) (2.0.1)  
  
[notice] A new release of pip is available: 23.3.1 -> 24.0  
[notice] To update, run: python.exe -m pip install --upgrade pip  
PS D:\SEM-6\Advanced Web Technology\Lab_11> python app.py  
* Serving Flask app 'app' (lazy loading)  
* Environment: production  
  WARNING: This is a development server. Do not use it in a production deployment.  
  Use a production WSGI server instead.  
* Debug mode: on  
* Restarting with stat  
* Debugger is active!  
* Debugger PIN: 868-049-390  
* Running on http://127.0.0.1:5000/ (Press CTRL+C to quit)  
127.0.0.1 - - [25/Apr/2024 22:23:59] "GET / HTTP/1.1" 200 -  
127.0.0.1 - - [25/Apr/2024 22:23:59] "GET /favicon.ico HTTP/1.1" 404 -
```



2. Connect the Flask Web-Application using SQLAlchemy to PostgreSQL and perform CRUD commands.

app.py

```
from flask import Flask, request, jsonify
from flask_sqlalchemy import SQLAlchemy

app = Flask(__name__)
app.config['SQLALCHEMY_DATABASE_URI'] = 'postgresql://postgres:password@localhost/user2'
app.config['SQLALCHEMY_TRACK_MODIFICATIONS'] = False
db = SQLAlchemy(app)

class User(db.Model):
    id = db.Column(db.Integer, primary_key=True)
    username = db.Column(db.String(80), nullable=False)
    email = db.Column(db.String(120), nullable=False)

    def __repr__(self):
        return f'<User {self.username}>'

@app.route('/users', methods=['GET'])
def get_users():
    users = User.query.all()
    return jsonify([{'id': user.id, 'username': user.username, 'email': user.email} for user in users])

@app.route('/users', methods=['POST'])
def create_user():
    data = request.json
    new_user = User(username=data['username'], email=data['email'])
    db.session.add(new_user)
    db.session.commit()
    return jsonify({'message': 'User created successfully'})
```

```

@app.route('/users/<int:user_id>', methods=['PUT'])
def update_user(user_id):
    user = User.query.get_or_404(user_id)

    data = request.json

    user.username = data.get('username', user.username)
    user.email = data.get('email', user.email)

    db.session.commit()

    return jsonify({'message': 'User updated successfully'})

@app.route('/users/<int:user_id>', methods=['DELETE'])
def delete_user(user_id):
    user = User.query.get_or_404(user_id)

    db.session.delete(user)

    db.session.commit()

    return jsonify({'message': 'User deleted successfully'})

if __name__ == '__main__':
    app.run(debug=True)

```

```

PS D:\SEM-6\Advanced Web Technology\flask_crud> python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 868-049-390

```

Create

```

PS D:\SEM-6\Advanced Web Technology\flask_crud> Invoke-WebRequest -Uri "http://localhost:5000/users" -Method Post -ContentType "application/json" -Body '{
>>   "username": "new_user",
>>   "email": "new_user@example.com"
>> }'
>>

message
-----
User created successfully

```

Update

```

PS D:\SEM-6\Advanced Web Technology\flask_crud> Invoke-WebRequest -Uri "http://localhost:5000/users/1" -Method Put -ContentType "application/json" -Body '{
>>   "username": "updated_username",
>>   "email": "updated_email@example.com"
>> }'
>>

message
-----
User updated successfully

```

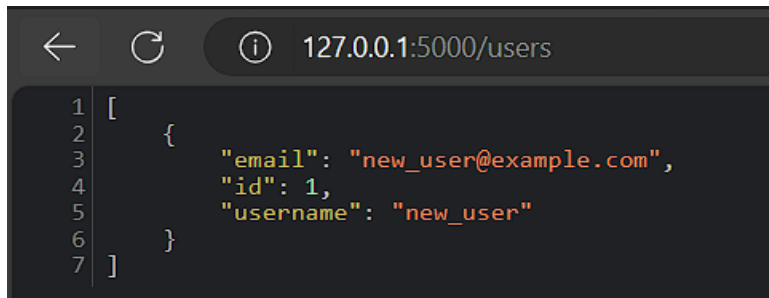
Delete

```
PS D:\SEM-6\Advanced Web Technology\flask_crud> Invoke-RestMethod -Uri "http://localhost:5000/users/1" -Method Delete -ContentType "application/json" -Body '{
>>   "username": "updated_username",
>>   "email": "updated_email@example.com"
>> }'
>>

message
-----
User deleted successfully
```

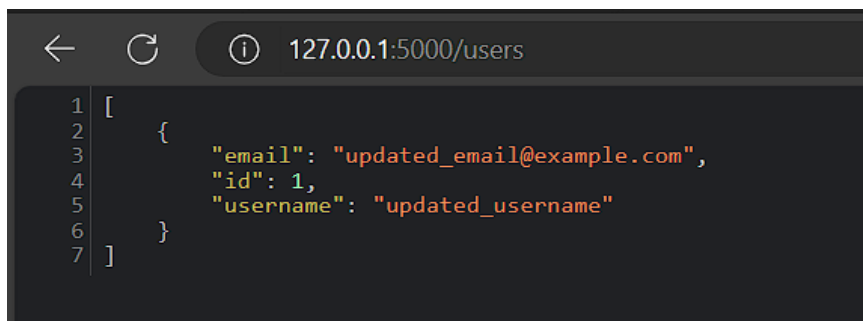
Output:

Create



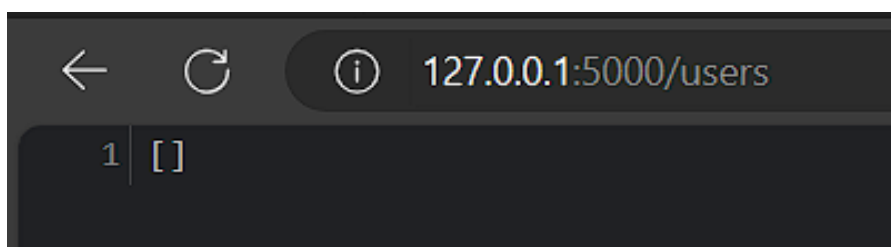
```
1 [
2   {
3     "email": "new_user@example.com",
4     "id": 1,
5     "username": "new_user"
6   }
7 ]
```

Update



```
1 [
2   {
3     "email": "updated_email@example.com",
4     "id": 1,
5     "username": "updated_username"
6   }
7 ]
```

Delete



```
1 []
```